

Agency Incentives and Disparate Revenue Collection: Evidence from Chicago Parking Tickets*

Elizabeth Luh[†]
SEC

Benjamin Pyle[‡]
Boston University

James Reeves[§]
University of Colorado Denver

April 2026

Abstract

We leverage a sharp 2012 parking fine increase for failing to purchase vehicle registration to examine disparate ticketing patterns across enforcement agencies in Chicago. Using an event-study framework, we find that Chicago police increased their enforcement of car registration non-compliance in Black relative to non-Black neighborhoods, with no observed disparate response for non-police enforcement agencies. This disparity is unexplained by compliance and is instead driven by departmental revenue incentives and multitasking-driven lower marginal search costs in Black neighborhoods. Disparate enforcement also exacerbated existing gaps in financial instability, generating increased rates of ticket non-payment and bankruptcy filings in Black neighborhoods.

*We thank Anne Burton, Aimee Chinn, Keith Finlay, Shannon Graham, Caroline Hoxby, Chinhui Juhn, Elaine Liu, Vikram Maheshri, Mike Makowsky, Steve Mello, Jon Moreno-Medina, Mike Mueller-Smith, Kerri Raissan, Sonja Starr, Allison Stashko, Gergely Ujhelyi, and Anastasia Wilson for helpful comments and suggestions. We also thank participants at Boston University’s Law and Economics Working Paper Seminar and Law Faculty Workshop, the 2024 Texas Economics of Crime Workshop, Vanderbilt University’s Law and Economics Seminar, Kennesaw State University, the 2024 Western Economic Association Meeting, University of Texas at San Antonio, and the 34th Annual Meeting of the American Law and Economics Association. A previous version of this paper was circulated as “Disparate Fine Collection: Evidence using Chicago Parking Tickets.” The Securities and Exchange Commission disclaims responsibility for any private publication or statement of any SEC employee or Commissioner. This article expresses the authors’ views and does not necessarily reflect those of the Commission, the Commissioners, or other members of the staff.

[†]Securities and Exchange Commission, United States, elizabethluh@gmail.com

[‡]School of Law, Boston University, United States, benpyle@bu.edu

[§]Department of Economics, University of Colorado Denver, United States, james.reeves@ucdenver.edu

1 Introduction

Parking fines are major revenue generators for United States cities. For example, in 2016, Chicago raised \$264 million from parking citations, equivalent to an annual \$97.20 per-capita tax; similarly, New York City raised \$565 million from parking fines in the year prior (Diskin, 2019; Digital Editors, 2021). The ability of a city to raise revenue from these types of fines depends on both its enforcement and collection policies and its residents’ ability to pay these fines. However, reliance on fines and fees as a local revenue stream can impose significant economic consequences on residents, especially those with lower ability to pay (Makowsky and Stratmann, 2009, 2011; U.S. Department of Justice Civil Rights Division, 2015).¹

Local governments can use many different types of agents to enforce parking violations, including police officers, parking enforcement aides, and other third-party contractors. The equity and efficiency of parking enforcement across neighborhoods are therefore dependent in part on the behavior of agents whom the government hires. Consequently, agents may disparately enforce violations across areas if they are maximizing a different objective function than the government - for example, minimizing search costs or focusing on alternative, non-parking enforcement responsibilities.² To the extent such wedges between government and agent exist and are correlated with racial and ethnic divisions within a metropolitan area, agents may respond to collection incentives in ways that generate disparate outcomes in the population while also harming revenue.

In this paper, we examine how an increase in motor vehicle registration cost, which we refer to as “stickers,” and the related fine for non-compliance affected ticket enforcement patterns in Chicago. This 2012 policy increased the cost of typical vehicle registration from \$75 to \$85 (\$120 to \$135 for larger vehicles), a 13% increase, and the fine for registration non-compliance from \$120–\$200, a 67% increase. Thus, it simultaneously made compliance more expensive and enforcement more profitable relative to other parking fines. Aggressive enforcement and punishment had severe consequences for Chicago residents, particularly in predominantly Black neighborhoods, leading to claims of disparate impact.³

Since both the Chicago Police Department (CPD) and other agents, primarily but not exclusively parking enforcement agents (henceforth, non-CPD agents), can enforce municipal

¹Propublica estimates that unpaid parking debt alone in Chicago totals over \$1.5 billion debt (Samuelson, 2018).

²There is a long literature examining multi-task principal-agent models and empirically investigating agent responses in the face of differing incentives. For example, see Holmstrom and Milgrom (1991) and Aghion and Tirole (1997) for the canonical theory, and Jacob and Levitt (2003) and Knutsson and Tyrefors (2022) for empirical work in the context of teachers and ambulances, respectively.

³While predominantly Black neighborhoods account for only 22% of tickets, they hold 40% of all ticket debt. (Sanchez and Kambhampati, 2018).

parking laws, we separately examine the impact of the sticker tax increase on both types of agents, with the major distinction being that while non-CPD agents are evaluated on their ticketing volume, CPD officers are not. Furthermore, CPD officers have additional, broader responsibilities to “work for the benefit of its citizens by protecting life and property from harm and maintain order” (Department of Human Resources, 2023). Thus, the budget reform provides a unique opportunity to evaluate how governments use different incentives across various agents to affect revenue-generating enforcement and the downstream impact this has across residents.

We test for disparate enforcement across the two types of agents using administrative parking ticket data from 2007 to 2016 in Chicago and a difference-in-differences (DiD) framework, estimating the relative change in sticker enforcement across Black and non-Black neighborhoods.⁴ Across Black versus non-Black neighborhoods, our results show consistent evidence of a disparate response for CPD-issued tickets. We find that CPD sticker enforcement increased by nearly 2,500 tickets in Black relative to non-Black neighborhoods per year. Conversely, non-CPD ticketing activity exhibits no such changes across neighborhoods.

The relative increase in sticker enforcement in Black neighborhoods also had sharply negative implications on the source of collected revenue and financial outcomes (e.g., bankruptcy) of those ticketed. At the ticket level, we find that revenue per ticket decreased by \$27 in Black neighborhoods relative to non-Black neighborhoods, indicating increased instances of nonpayment and financial strain. Accordingly, we estimate that more than half of the additionally issued sticker tickets resulted in notices of nonpayment, with a further 10% resulting in bankruptcy filings.⁵ Despite these low payment rates, on aggregate, the increased sticker enforcement shifted the tax burden from non-Black neighborhoods to Black neighborhoods. Relative to non-Black neighborhoods, annual collected revenue increased by over \$270,000 more in Black neighborhoods, representing a regressive increase of \$5 per capita. Conversely, non-CPD agents collect greater revenue from non-Black neighborhoods.

Given that both the cost of registration and the fine increased simultaneously, these disparities could reflect differential enforcement or compliance across Black and non-Black neighborhoods.⁶ Using sticker purchasing data at the neighborhood level from 2008–2016, we rule out changing compliance, as we find no differential changes in purchasing behavior

⁴We also stratify across income and ability to pay and do not find similar patterns of disparate enforcement for CPD and non-CPD issued parking tickets.

⁵Over our sample period, the number of Chapter 13 bankruptcies with parking debt rose substantially, from 1,000 per year to 10,000 per year. Similarly, the median parking debt per case more than doubled, from \$1,500 to \$3,900 (Sanchez and Kambhampati, 2018).

⁶Variation in neighborhood characteristics (e.g., number of parking meters) and its residents (e.g., number of vehicle owners with valid registration) may also lead to differential non-compliance which could warrant differential enforcement, holding fixed policing patterns.

across neighborhoods. Through a complementary decomposition exercise, we further document outsized differential enforcement in Black neighborhoods, even after accounting for differences in the number of unregistered vehicles by neighborhood. Coupled with similar purchasing responses, these results further underscore that the patterns we document are predominantly due to changes in CPD enforcement rather than simply reflecting divergence in residents' compliance with the policy.

What drives the differences in ticketing behavior across enforcement agencies? We consider two complementary potential mechanisms, finding evidence in favor of each. First, we find strong evidence that the disparate enforcement stems partly from multitasking responsibilities due to crime responses. Specifically, we show that the increase in sticker ticketing is strongest in areas with the highest crime rates, suggesting that CPD officers are increasing their ticketing effort in areas where their search costs are lowest. Moreover, we distinguish between multitask ticketing which occurs in the immediate vicinity of a criminal incident and multitask ticketing which occurs due to greater time spent in high-crime areas. The vast majority of the disparate ticketing responses we document are due to the latter channel. In sharp contrast, non-CPD enforcement is independent of local crime rates, as such multitasking concerns are not present.

Second, we find contextual and empirical evidence that CPD faces top-down revenue collection pressures. Local officials view ticketing revenue as a practical method for revenue generation (Lentino, 2017), and memos from City Hall also raise concerns when collected ticket revenue falls below projected targets (Rivera, 2010). Empirically, we show that on the first day sticker enforcement is legally permitted, CPD officers increase their ticketing volume by nearly 50% post-reform, compared to the pre-reform level. On the other hand, non-CPD agents face performance incentives based on ticket volume, suggesting their behavior should be independent of changes to fine amounts. Accordingly, we find little evidence that non-CPD agents exhibit the same change in ticketing dynamics when sticker tickets are permissible. Moreover, in a complementary exercise leveraging the privatization of Chicago's parking meters, we show that CPD enforcement of expired meters, relative to non-CPD enforcement, falls sharply after the rights to ticketing revenue were sold to a private investment group. This response is also consistent with CPD officers altering their enforcement behavior in response to the destination of the ticketing revenue, as post-privatization, revenue accrues to the investment group rather than the city. Taken together, we view multitasking as the key distortion to CPD behavior and the top-down revenue collection incentives as the catalyst that drives the disparate and distinct enforcement patterns.

Finally, we show that disparate ticketing in response to the reform is a widespread phenomenon and not confined to a handful of "bad apple" officers, consistent with task structure

and departmental coordination as key drivers of officer behavior. We estimate CPD officer-specific responses to the budgetary reform and show that 65-85% of officers increase their sticker ticketing activity after the fine increase. Across the officer distribution, the marginal sticker ticket is also almost twice as likely to be written in a Black neighborhood than in a non-Black neighborhood. Regressing officer characteristics against these estimated policy responses reveals few strong correlations, but the empirical patterns we observe can also not be fully explained by the neighborhood demographics of officer assignments. Instead, we conclude that the disparate enforcement across neighborhoods was part of a broader departmental phenomenon and revenue collection effort in response to the budget reform and existing deficit.

This paper builds empirical evidence on incentives and the behavior of public sector agents and their role in revenue generation for local governments.⁷ We build on prior work studying the responses of police officers to pay (Mas, 2006) and opportunities for overtime compensation (Chalfin and Goncalves, 2023). We show that police officers in our setting are responsive to governmental revenue goals in ways that are not present among privately contracted agents, likely due to differences in incentives across agencies, and that this response leads to disparate revenue collection and financial outcomes in the population.⁸

In doing so, we connect to a rich literature in public finance studying how the structure of tax enforcement and laws may contribute to disparities and evasion (Allingham and Sandmo, 1972; Slemrod and Yitzhaki, 2002). For instance, work has shown that Black homeowners pay higher property taxes (Avenancio-León and Howard, 2022), Black taxpayers are more likely to be audited for claiming the Earned Income Tax Credit (Elzayn et al., 2025), and Black married couples are more likely to incur a “marriage penalty” in income taxes (Brown, 2022; Holtzblatt et al., 2023). We highlight an additional channel through which revenue-generating policies have disparate and regressive impacts: the enforcement of a tax on car ownership through city car registration.

Previous work has also studied the same policy variation to understand bankruptcy behavior (Morrison, Pang, and Uettwiller, 2020). Using a similar identification strategy to what we will show, Morrison, Pang, and Uettwiller (2020) document that Chicago’s ticket enforcement policy change dramatically increased Chapter 13 bankruptcy filings, particularly in Black communities. They note that the drivers of Chapter 13 bankruptcy cases were not

⁷For theoretical work in this area, see Prendergast (2007, 2008) and for a review on incentives and decision making, see Kamenica (2012).

⁸Our results also build on a recent strand of work studying the consequences of both multitasking (Reeves 2024) and specialization (Dee and Pyne, 2022; Davis et al., 2025) among police agencies. Additionally, prior work in public finance shows tax collectors are responsive to performance pay (Kahn, Silva, and Ziliak, 2001; Khan, Khwaja, and Olken, 2016). For more on the role of law enforcement in revenue collection, see Harvey (2020), Goldstein, Sances, and You (2020), and Makowsky, Stratmann, and Tabarrok (2019).

lawyer-led but driven by Chicago’s overall enforcement and collection decisions. We add to this work by showing that the upstream enforcement decisions were due to sticker ticket enforcement by the CPD and not other enforcement agencies.

Finally, our study links disparate policing and the actions of government agents to financial fragility that disproportionately falls along racial lines.⁹ In contrast to prior work which emphasizes disparate policing in highway patrol agencies (e.g., Goncalves and Mello, 2021), our focus on non-moving violations and metropolitan policing implies that officers’ ticketing choices are less likely to be confounded with concerns for public safety. For example, failing to stop a speeding driver in the case of highway ticketing choice could have more severe consequences than failing to ticket an improperly parked car. Thus, Chicago police are effectively being tasked as tax collectors. Furthermore, given our rich data, we can cleanly measure the monetary and revenue implications of disparate (tax) policing.¹⁰

2 Institutional Background and Setting

Chicago is often recognized as one of the most spatially segregated cities in the United States and has wide racial gaps in socioeconomic outcomes. For example, Black-owned businesses are valued at less than one-twelfth the value of white businesses, and the median household income for Black families (\$30,303) is less than half that of white families (\$70,960) (ProsperityNow, 2017). These inequalities also extend to local crime rates and rates of contact with the criminal justice system. Black neighborhoods experience substantially higher homicide rates (Sharkey and Marsteller, 2022), and Black motorists comprise the majority (60%) of traffic stops in the city (Goudie et al., 2024). Importantly for this paper, this increased exposure to law enforcement also raises the capacity for greater sticker ticket enforcement, as officers are more likely to observe any expired registrations given the higher rates of contact in Black neighborhoods. These features imply that any changes to CPD sticker enforcement efforts will also have a disproportionate impact in Black neighborhoods.

The city of Chicago relies heavily on parking ticket revenue, with 7% of its 3.6 billion dollar operating budget coming from the fines it collects. Each year, the city issues over 3 million tickets for parking violations, vehicle compliance, and automated traffic camera violations (Sanchez and Kambhampati, 2018).

⁹See Owens and Ba (2021) for a comprehensive literature review on policing and disproportionate burden across demographic subgroups.

¹⁰Recent work on financial sanctions in the justice system has found mixed evidence of the impact of these sanctions on outcomes. Pager et al. (2022) and Finlay et al. (2024) show that financial sanctions from a criminal conviction have no long-term or short-term impact on labor market or recidivism outcomes. However, Goncalves and Mello (Forthcoming), Mello (2025), Kessler (2020), and Hansen (2015) have found negative impacts on financial outcomes but reduced recidivism from harsher fines.

Chicago's parking fines are also particularly punitive. Unpaid fines double 25 days from the initial ticket issue date and are subject to additional late fees. If a driver has repeated violations or fails to pay their fines, they can receive a vehicle seizure notice (eventually having their car seized) and ultimately have their license suspended.¹¹ Drivers can choose to enter into a payment plan with the city, but to qualify for the standard payment plan, the driver must pay a \$1,000 down payment on total vehicle debt plus payment in full on any tow, boot, or storage fees. If the driver is unable to commit to a plan, they can declare bankruptcy. Chicago also has anti-scofflaw rules that prevent those with unpaid tickets or debts to the city from accessing contracts, licenses, or grants. For example, municipal jobs, such as teaching in a classroom, are inaccessible for those with unpaid parking tickets. With no statute of limitations, parking debt in Chicago can follow the driver for an entire lifetime.¹²

Column 1 of Table 1 shows the top ten parking tickets by total revenue collected from 2007 to 2011. The most issued ticket has a fine of \$50. Of the top tickets, only three are related to parking (residential permit parking, expired meter, parking in prohibited areas). Four of the most popular tickets relate to having correct licensing or registration.

The most punitive parking fine is for failing to properly display the city sticker on the car windshield. The city sticker, which is colloquially known as the 'sticker tax,' is an annual registration fee that Chicago residents with vehicles must pay to own a vehicle in the city. While the registration is \$75 for sedans and \$120 for larger passenger vehicles, the fine for failing to buy the sticker or failing to display the sticker is almost double that of the next most expensive ticket, at \$120 plus a \$40 late fee. In October 2011, the city announced it would be raising the registration fee for the stickers from \$75 to \$85 for smaller passenger vehicles (<4,500 lbs) and \$120 to \$135 for larger passenger vehicles (Civic Federation, 2012). Further, the fine for not paying the tax would increase by two-thirds, from \$120 to \$200, effective February 2012.

The fee increase was announced in conjunction with other aggressive revenue-generating policies in an attempt to close a \$637 million projected deficit in the 2012 budget (Emanuel, 2011). One of these policies was an aggressive debt collection plan that directly affected how the city collected and enforced payment of parking ticket fees. This plan would allow the city to begin garnishing the wages and tax returns for high debtors in addition to suspending

¹¹Specifically, after three unpaid parking tickets, red light tickets, or speed camera tickets within a year or two unpaid parking tickets, red light camera tickets, or speed enforcement tickets that are one year past due, the car can be impounded or booted, and the vehicle owner receives a seizure notice. After ten or more non-moving violations or five unpaid tickets from automated red-light or speed cameras, the city of Chicago suspends the vehicle owner's driver's license.

¹²In contrast, Los Angeles and New York City have statutes of limitations that are 5 and 8 years, respectively.

licenses and impounding cars. Once the maximum amount of fees had been levied, the city could garnish drivers’ state tax returns and 15% of wages (Andriesen, 2012). The stated goal of this aggressive debt collection plan for parking and traffic infractions was to reduce employee indebtedness and to hold rental car companies accountable for their parking fines. For city employees, the mayor announced additional punishments. For example, City Hall workers could face suspension or be fired for owing anywhere between \$250 to \$1000 and more (Ruthhart and Reporter, 2014).

Both the Chicago police and parking enforcement agents (succinctly, non-CPD) can issue parking citations. Non-CPD agents can enforce non-moving ordinances in Chicago and are either employed directly by the city (e.g., the Department of Revenue) or through a third-party firm (e.g., SERCO). Similar to other occupations with relatively narrow objective functions (e.g., Lazear 2000), non-CPD agents believe they are evaluated based on measurable output, such as their ticketing productivity and are sometimes promoted based on their tickets per shift. Indeed, non-CPD agents publicly claim via work testimonials that the “work hours are late in extreme weather conditions all they care about is their ticket counts” and “all they care about is a high ticket count.” We are unaware of alternative evaluation schemes that link non-CPD agent incentives to downstream ticket outcomes (e.g., payment probabilities) or ticket revenue collected. Based on this anecdotal evidence, we believe that modeling non-CPD agents as ticket volume maximizers is therefore a reasonable approach.¹³

In contrast, Chicago police officers’ main job function is not parking enforcement, and their parking ticket productivity is not as important to their careers.¹⁴ However, ticketing revenue implicitly affects the operations of the Chicago Police Department as budget deficits can lead to citywide hiring freezes and potential layoffs (Spielman and Schuba, 2024). In addition, local officials view ticketing as a practical avenue for revenue generation (Lentino, 2017) and memos from City Hall were written when CPD-collected ticket revenue fell below targets. Subsequently CPD worked closely with the Revenue Department to track ticketing productivity (Rivera, 2010). Thus, while CPD officers may not explicitly face the same type of productivity-based incentives as non-CPD agents, they nonetheless still face top-down pressure for ticketing as a source of revenue generation. We present a simple rationalizing model that formalizes the differences between CPD and non-CPD objectives in Appendix

¹³This information supporting our understanding of the non-CPD agents’ objective function is based on reading work testimonials from Indeed.com, see <https://perma.cc/9DPN-QXF4>.

¹⁴Furthermore, Chicago policymakers have been shifting toward banning traffic ticket quotas for police officers in recent years. In 2019, Illinois passed a law explicitly forbidding law enforcement agencies from evaluating personnel based on their ticket-issuing productivity. This was partly due to the criticism towards CPD for mandating a minimum number of traffic stops prior to the passage of the law (Main, 2017). To the best of our knowledge, CPD officers were not required to fulfill any quotas for parking tickets and any reference to ticket quotas was in relation to traffic infractions.

B.

3 Data

ProPublica Illinois, in partnership with WBEZ Chicago, obtained and publicly released parking ticket data from the city of Chicago from 2007 to 2018. The data include information on the date and time of the ticket, where the vehicle was parked, the badge number of the ticketing officer, de-identified license plates, make, registration zip code, citation reason, and, importantly, payment status. This payment status includes information on the ticket outcome, such as whether the vehicle owner received a notice of seizure, whether the vehicle owner received a notice of driver’s license suspension, and whether the vehicle owner declared bankruptcy as a result of the ticket. It also includes information on how much of the fine remains unpaid, the date of the last payment, and the initial fine amount.

A key object of analysis for this paper is the area or neighborhood where the ticket was given. We consider two levels of aggregation: zip code and census tract, the former of which serves as our primary definition of neighborhood and the latter of which we use for robustness checks. The raw ticketing data does not include zip code or tract of violation location. To recover this information, we use the Census Geocoder, which successfully maps street addresses to zip codes for a vast majority of the tickets in our sample. The remainder are coded using geocoder.us. Where zip codes are unavailable from these matches, we use their latitude and longitude to map into the relevant zip code based on Census GIS shapefiles. To determine if a zip code neighborhood is considered Black or non-Black, we match each zip code to the Census 5-digit Zip Code Tabulation Area (ZCTA5) using the American Community Survey (ACS) 5-Year estimates for 2007 - 2011 (Manson et al., 2024). The ZCTA5 are approximate area representations of zip codes in the United States but are not exact matches for zip codes since ZCTA5s are aggregated from Census blocks. Despite this, they are close matches for each other. Tract demographics can be measured directly. We show in Section 5.1 that our results are robust to our choice of geographic aggregation.

Additionally, we obtained sticker registration (purchase) data by zip code using a Freedom of Information Act request filed with the City Clerk of the city of Chicago. While ideally we would be able to measure compliance at the vehicle or individual level of observation, to protect the purchasers’ privacy, this information only contained the zip code of the buyer’s billing address. The data contains information on the date and time of the purchase, the full purchase amount, and the type of vehicle the sticker was for. We map each sticker purchase to a ZCTA5 using the zip code.

3.1 Ticketing Sources of Revenue and Descriptive Statistics

We present summary statistics describing the ticketing data in Table 1.¹⁵ This table reports average annual characteristics of the top ten revenue-generating tickets from 2007-2011 and characteristics for the same set of tickets from 2012-2016.¹⁶ The type of ticket is listed in each row. We show annual ticket volume, annual ticket revenue, modal fine amount, the revenue share of the listed ticket along with the revenue share rank in Columns 5 and 11, and the ratio of revenue received to expected revenue, where expected revenue is calculated as the base fine amount times ticket volume. The revenue share is less than one when collected revenue plus applicable late fees is less than the expected collected amount if all written tickets were paid on time and is greater than one if the collected revenue plus applicable late fees exceeds this expectation. While sticker tickets were among the top two revenue-generating tickets prior to the policy, they are far and away the highest revenue-generating ticket in the post-reform period, collecting around 50 percent more revenue than the next most profitable ticket. This increase in revenue is despite a sharp decline in the ratio of revenue to expected revenue, suggesting lower payment rates and increases in financial strain among the ticketed population.

Our analysis focuses on comparisons between Black (defined as $\geq 75\%$ Black) neighborhoods and non-Black neighborhoods across several ticket-related outcomes, including the type of ticket based on the violation code (sticker ticket or another type of ticket). We also include several point-in-time measures (these measures evolve over time but are current as of the latest data extract in 2019): the amount of revenue collected from the ticket, if a ticket was dismissed (either internally or as a result of an appeal) fully paid, given a notice of non-payment (if the ticket was not yet paid and the city sent a notice to the address on file for that vehicle), or included as a debt in a consumer bankruptcy case.

Our choice to define neighborhoods as predominantly Black if their population share is at least 75% is based on the distribution of neighborhood demographics in Chicago, which we illustrate in Appendix Figure A1. There is a clear bimodal distribution of the population demographics across neighborhoods, which are largely Black or non-Black, with a missing middle. In robustness checks below, we show that our results are remarkably stable regardless of the exact threshold we use or when we exclude neighborhoods in the middle of the distribution to make even starker cross-neighborhood comparisons.

We show descriptive statistics summarizing the most salient data features in Table 2

¹⁵Displayed separately by neighborhood demographic group in Appendix Table A1.

¹⁶We limit the data to 2016 to ensure our estimates are uncontaminated by a series of Chicago Police Department Reforms (e.g., <https://www.chicago.gov/content/dam/city/depts/mayor/Public%20Safety%20Reforms/ProgressReport-PoliceReforms.pdf>) enacted after 2016. Our results throughout are highly similar if we use data through 2018 instead.

for the five years before the sticker policy change (2007-2011). Panel A of this table shows information at the ticket level. About 60% of tickets in Black neighborhoods are written by CPD, who also write fewer than half of tickets in non-Black areas. About 15% of tickets written in Black areas are sticker tickets, while this rate is close to 6% in non-Black areas. Outcomes after receiving a sticker ticket are worse for people who receive tickets in Black areas as they are less likely to fully pay the ticket or have the ticket dismissed and more likely to be involved in bankruptcy or have a non-payment notice.

Panels B and C show these same outcomes at the neighborhood level. Black areas have slightly larger populations on average (about 51 thousand compared to about 47 thousand in non-Black areas), but also have fewer vehicles (about 17 thousand compared to 21 thousand). Black neighborhoods have fewer stickers purchased (16 thousand compared to 21 thousand) and slightly lower ratios of stickers purchased relative to the number of total vehicles (95% compared to 100%).¹⁷ The number of total tickets written by the CPD is lower in Black neighborhoods, although more of them are sticker tickets. Substantially more sticker tickets are written in Black neighborhoods (2000 more per year).

We present the raw geographic dispersion of several key measures in Appendix Figure A2. This figure shows which zip codes are greater than 75% Black (no zip code crosses the threshold to change classifications during our sample period). It also shows the dispersion of crime (from the CPD open data portal), sticker tickets issued by CPD and non-CPD agents, and parking lots and garages in the city of Chicago (which we code based on data from Open Street Maps). The similarities between the geographies of crime and CPD-issued sticker tickets provide early evidence of the multi-tasking and effort channels discussed throughout this paper. Moreover, both CPD and non-CPD agents issue tickets throughout the city, suggesting our results below cannot be rationalized under alternative theories where non-CPD only work in non-Black areas or where Chicago police are not present.

4 Empirical Strategy

Our objective is to estimate the relative effects of the ticketing reform on parking enforcement patterns across neighborhoods. We summarize this difference-in-differences approach in two equations. First, in Equation (1), we present event study evidence comparing majority ($\geq 75\%$) Black ($Black_i$) areas to non-majority Black areas, where the α_t coefficients trace the evolution of an outcome of interest relative to 2011, the year prior to the sticker policy change. This allows us to carefully evaluate parallel trends, a requisite identifying assumption.

¹⁷Our measure of neighborhood-level vehicles is survey-based and constructed from the aggregation of several categories of survey responses, one of which is top-coded. Therefore, a strict interpretation of the number of vehicles should be made with caution.

Another identifying assumption for a difference-in-differences analysis is that the treatment has no impact on the control group. In our setting, changing the sticker policy may also impact outcomes in non-Black areas. What we recover from our analysis is not the overall impact of the policy on our outcomes but rather the differential impact of the policy on Black areas relative to their non-Black counterparts. Formally, for neighborhood i and year t , we estimate:

$$Y_{it} = \alpha_0 + \sum_{\substack{t=2007 \\ t \neq 2011}}^{2016} \alpha_t (Black_i \times Year_t) + \sigma_i + \lambda_t + \varepsilon_{it} \quad (1)$$

Notably, our λ_t fixed effects control for any common temporal shocks and our σ_i fixed effects control for any baseline differences across neighborhoods, such as differences in policing levels.¹⁸ Even in the presence of such baseline differences, our results can be interpreted as capturing changes in police ticketing or taxpayer compliance behaviors, holding fixed the pre-reform neighborhood policing concentration and registration compliance levels. Second, we summarize our results in a single point estimate in Equation (2). Conditional on our identifying assumptions, β_1 of this equation summarizes the impact of the policy change in the follow-up period (2012-2016).

$$Y_{it} = \beta_0 + \beta_1 (Black_i \times Post_t) + \sigma_i + \lambda_t + \epsilon_{it} \quad (2)$$

A two-way fixed effects strategy is appropriate in this setting as treatment occurs simultaneously for all treated units, allowing us to avoid concerns of negative weights in the presence of treatment heterogeneity (Goodman-Bacon, 2021). Additionally, we handle the potential complications of a continuous treatment by defining a binary treatment indicator defined as ($\geq 75\%$) Black zip codes (Callaway, Goodman-Bacon, and Sant’Anna, 2024).

Our identification strategy is most credibly interpreted as generating partial equilibrium results. However, given stable and flat trends in ticketing behavior in white neighborhoods and the additional observation of little change in non-CPD behavior, the approach may credibly inform more general conclusions.

Sticker purchasing data is only available at the zip code level while ticketing data is available at both the zip and tract geographic levels. For internal consistency across exercises, we show the zip code results in the main text. In robustness checks below, we show that our results are similar when we instead define neighborhoods at the tract level. Moreover, while discretizing the treatment allows us to avoid the challenges of continuous treatment in

¹⁸Indeed, using staffing data from the Invisible Institute, we estimate that Chicago Police Districts that predominantly serve Black neighborhoods have around 4% more police officers in the years prior to the reform.

DiD settings, the choice of treatment cutoff may be important for interpreting our results. Section 5.1 shows that our results are robust to both higher and lower thresholds.

5 Estimates of Disparate Ticketing and Outcomes

We begin our analysis by examining ticketing trends in the raw data. Panel A of Figure 1 plots the yearly number of CPD-issued sticker tickets issued in Black and non-Black neighborhoods (blue and gray, respectively), both in level and per-capita terms (solid and dashed lines, respectively). Panel B, plots the same series for non-CPD agencies. In Panel A (CPD), prior to the 2012 reform, both sticker ticket series were trending downwards, with lower ticket volumes year-over-year. After 2012, however, ticket volumes in Black neighborhoods exhibited a precipitous jump upward, exceeding their pre-policy levels, while ticket volumes in non-Black neighborhoods largely flattened. In contrast, the series for non-CPD agencies (Panel B) are relatively flat, displaying little noticeable changes pre- or post-reform.¹⁹ Together, these raw data series depict evidence that law enforcement agencies enforced sticker ticketing more in Black neighborhoods, compared to non-Black neighborhoods.

In order to causally link the disparate enforcement to the change in sticker ticket volume and other outcomes, we must rule out other potential confounding factors. For example, neighborhoods may differ in their baseline financial strain, resulting in differential non-compliance or increased probabilities of non-payment. Neighborhoods may also experience differential policing patterns, which increase the likelihood that any non-compliance is noted by law enforcement. Our empirical strategy accounts for time-invariant differences across neighborhoods, along with common temporal shocks, although the existence of remaining time-varying differences across neighborhoods would violate the parallel trends assumption and undermine any causal interpretation.

In Figure 2, we report event-study coefficients on the interactions of neighborhood type and year indicators from Equation (1). In the plot region, we also report the summary difference-in-differences calculation, estimated using Equation (2). We begin in Panel A by examining changes in sticker ticketing volume around the time of the policy, separately for CPD and non-CPD agents. Consistent with the patterns in the raw data, we find sharp relative increases in sticker ticketing in Black neighborhoods for CPD officers (blue series). On average over the follow-up period, this increase corresponds to 2,490 additional sticker tickets per year. Reassuringly, we also find little evidence of differential pre-trend violations,

¹⁹These opposing pre-reform trends across agencies are likely due to a gradual shift towards reliance on third-party contractors, rather than CPD officers for non-moving violation ticketing. Moreover, it's unlikely these patterns are driven by systematic moves of officers across districts as 68% of CPD officers do not move across districts during their tenure (Ba et al., 2022).

lending credibility to our research design. Despite the sharp increases in relative ticketing among CPD officers, we detect no such change for non-CPD agents, and if anything, find a small decrease in relative sticker ticket volume in Black neighborhoods.

Alongside the increase in sticker ticket volume, we also find a sharp rise in relative ticket revenue collected from Black neighborhoods in Panel B. On average, CPD ticketing behavior generates an additional \$270,000 per year from Black neighborhoods, which given the historical patterns of income and wealth in the city, represent a likely increase in tax regressivity. Similar to our findings in Panel A, we also detect no such relative increase in revenue from non-CPD agents.

A unique feature of our dataset is the ability to link tickets to their downstream outcomes, which we assess in Panels C through F. Among the nearly 2,500 additional annual sticker tickets in Black neighborhoods, we find that under 15% result in on-time payment (Panel C) and over half receive a notice of late payment (Panel D).²⁰ Perhaps strikingly, almost 10% of the sticker tickets are included in bankruptcy filings. This result on relative bankruptcy is consistent with the findings in Morrison, Pang, and Uettwiller (2020), where discharging debt through Chapter 13 bankruptcy filings permit individuals to retain access to their car and driver’s license, a benefit which accrues more strongly to Black residents. This rise in bankruptcy filings may also reflect differences in underlying financial strain in the population, where the marginal sticker ticket has a disparate impact on the financial health of Black versus non-Black residents. Finally, we observe some rise in dismissed sticker tickets in Black neighborhoods, which may be due to ex-post changes in dismissal policy and debt relief programs (e.g., Sanchez and Ramos, 2015) or a higher share of erroneous tickets. However, these increases in bankruptcy filings and dismissed tickets have important implications for the amount of *net* revenue raised. After accounting for the government’s cost for each of these types of hearing, we estimate that the additional revenue raised in Panel B in fact reflects a net loss of around \$230,000.²¹

It’s possible some of the observed disparity is due to changes in neighborhood differences in residents’ ability to pay for the sticker itself, resulting in differential non-compliance with the policy. Consequently, individuals who are unable to initially afford the sticker will also be unresponsive to the purchasing incentive induced by the sticker fine increase or be priced out because of the increase in the sticker price itself. Recall that our DiD estimates account for the initial level disparity across neighborhood types, and thus capture only the differential change in sticker ticketing frequency across neighborhoods, before and after the reform.

²⁰Tickets can also accrue revenue through late penalties and interest, even if they ultimately end up in bankruptcy or have a late-payment notice in our data.

²¹Specifically, we estimate the administrative cost of each bankruptcy hearing as \$2,000 (Elias, 2007; Mahoney, 2015) and each dismissed ticket as \$56 (National Center for State Courts, 2005).

However, we take seriously the idea of quantifying differential compliance and its interaction with departmental incentives as a mechanism for our results and discuss this point in detail in Section 5.2.

Finally, in Appendix Figure A3 we re-estimate Equation (1) at the ticket-level. The marginal CPD-written ticket is 3.8 percentage points more likely to be a sticker ticket. Conditional on a sticker ticket, the marginal ticket collects \$27 fewer in revenue in Black neighborhoods, is more likely to end in a notice of non-payment and end in bankruptcy. These results highlight how revenue-driven law enforcement efforts can be less fruitful revenue streams than otherwise forecasted. While the City’s partial equilibrium revenue forecasts which motivated the policy shift were based on assuming a 100% payment rate (Ramos and Sanchez, 2018), our results instead highlight how marginal increases to civil sanctions can ultimately harm both revenue and the local residents when the enforcement burden falls on economically disadvantaged populations.

5.1 Robustness Checks

Before assessing the mechanisms underlying our results, we first examine the stability of our estimates to a range of robustness checks in Appendix Table A2. Column 1 in each table reports the DiD estimate for each of our six main outcomes using only the raw data. Column 2 reproduces our estimate from the main text, adding zip code and year fixed effects in a standard DiD specification, although the results are little changed with these additions. Column 3 shows that our estimates are virtually unchanged when using the estimator suggested by Callaway and Sant’Anna (2021). Finally, in the last two columns, we use alternative cutoffs for defining zip codes as primarily Black. Changing the threshold to either at least 50% or 90% Black, rather than our baseline 75%, does not meaningfully affect our findings.²²

We further probe the empirical consequences of our 75% threshold to define Black neighborhoods in Appendix Figure A5. In this figure, we set the treated threshold at 5 percent increments (k) between 60% and 95% and exclude neighborhoods that have neither Black neighborhood shares above k percent nor shares below $1 - k$ percent. This restriction effectively drops the middle of the distribution and enables even starker cross-neighborhood comparisons. We report summary difference-in-differences estimates for our main outcomes in Panels A-F. Across all alternative thresholds, and both CPD and non-CPD departments, our results are remarkably similar to our main specification. Taken together with the natural alternative cutoffs we examined in Appendix Table A2, we conclude that the exact sample

²²Appendix Figure A4 finds broadly similar results but replaces the zip code fixed effects with tract fixed effects and redefines neighborhoods as majority-Black at the tract level using the same 75% threshold.

composition and cutoff chosen do not meaningfully affect our empirical results.

We also decompose the main analysis for the subset of tickets that have owner characteristics in Appendix Table A3 to determine whether the disparate ticketing patterns documented above largely accrue to individuals whose home neighborhood matches the racial composition of the ticketing zip code or if the results we find largely reflect commuter traffic instead. The first row of each panel (“main”) reproduces the difference-in-differences estimates from Figure 2. We then replace each outcome Y_{it} with $Y_{it} \times (Black Owner_i)$ in the second row and $Y_{it} \times (1 - Black Owner_i)$ in the third row, effectively decomposing the differential outcomes in Black neighborhoods to drivers from Black and non-Black neighborhoods, respectively.²³ Across nearly all outcomes, regressions, and ticketing agencies, we find that the burden of the disparate ticketing patterns in Black neighborhoods tends to fall on owners who are also from Black neighborhoods.

Finally, a threat to any event study identification strategy is the existence of concurrent policy changes that also differentially affect treated units. In our setting, Chicago changed their reporting policy to funnel reports for minor crimes (e.g., theft under \$500 and lost property) to the 311 calls for service line, rather than the 911 dispatch system. Although this shift occurred in approximately March 2011, which was the year prior to the policy variation we study, it may still have shifted policing patterns in a way that affects our results. While data constraints prevent us from formally examining differences in all 311 calls in our event-study framework, we nonetheless provide two pieces of evidence in Appendix Figure A6 that suggest this reform did not meaningfully affect policing patterns during our sample period.²⁴ In Panel A, we plot per 1,000 population rates of thefts under \$500 reported to the Chicago Police Department. If these calls were strategically diverted away from CPD and to the 311 line, we would see a sharp decline in this series after 2011. In fact, we find little evidence of any systematic changes, either in aggregate or differentially across neighborhoods. Moreover, in Panel B, we explore to what extent this routing policy may have shifted other types of 311 calls, again finding little evidence of any differential shifts. Taken together, we conclude that the 311 reporting change does not meaningfully affect policing patterns in a way that would affect the interpretation of our results.

²³We note that since not all of our tickets contain owner address information, the sum of the two disaggregated point estimates need not add up to the main results.

²⁴We were able to obtain historical 311 call data beginning in 2011 for complaints about abandoned vehicles, abandoned buildings, alley lights out, garbage carts, graffiti removal, pot hole reports, rodent baiting, sanitation code complaints, street lights (all or one out), tree debris, and tree trimming. For internal consistency across these exercises, we plot the raw time series across neighborhoods.

5.2 Mechanisms: Search Costs (Not Compliance) Drive Results

CPD and non-CPD agents perform different tasks and face different incentives. CPD officers have a broad set of responsibilities, including public safety priorities along with ticketing, while non-CPD agents are more specialized in ticketing. While the two groups seem to have reached an equilibrium prior to the ticketing reform, the incentives and responsibilities faced by the two agencies imply, and we empirically find, differing responses to the reform, with CPD agents responding more strongly to these incentives (see Appendix B for a formalized model). This subsection explores two potential explanations for the varied responses: changes in agency enforcement patterns driven by revenue considerations and resident compliance and evasion, finding support for the former via differential marginal search costs from multitasking and no evidence of the latter.

Departmental Incentives and Search Costs: One possible explanation of the different responses across ticket-writing departments is that the underlying performance evaluation scheme induces differential responses to the policy. Since non-CPD agents are evaluated on their ticket volume, their ticket-writing behavior was already maximizing ticket volume while minimizing search costs. Changing the revenue from a given ticket leaves the volume problem unchanged. In contrast, CPD officers face no such volume-based incentives to our knowledge and care more about revenue.²⁵ Thus, post policy change, the marginal benefit of writing an additional sticker ticket, from a revenue collection standpoint, has increased for CPD only. As a result, officers may induce greater search efforts into finding or ticketing vehicles without appropriate city stickers while multitasking on their other responsibilities.

There are two distinct ways in which CPD officers may adjust their behavior due to multitasking responsibilities. The first, which we term *passive multitasking*, implies that officers increase their ticketing-writing effort in high-crime areas where they are already patrolling, but largely during slack times. To build evidence for this channel, we first show that areas with higher crime levels experience greater increases in sticker ticketing.²⁶ In Panel A of Figure 3, we plot the joint distribution of changes in CPD-issued sticker tickets against the

²⁵See Section 2 for additional discussion on ticket volume and revenue incentives across agencies.

²⁶Reassuringly, we also find no differential changes in crime or police activity across neighborhoods around the time of the reform (see Appendix Figure A7). For all exercises leveraging Chicago crime data, we use publicly available data from the City of Chicago Open Data Portal, downloaded in September 2023 (see https://data.cityofchicago.org/Public-Safety/Crimes-2001-to-Present/ijzp-q8t2/about_data). We use the provided latitude and longitude to map incidents into zip codes and drop the small handful of incidents that we are unable to map, otherwise including all incidents in the data. To classify incidents into major crime categories, we apply the machine learning algorithm from Choi et al. (2023) to each incident's primary type field.

average annual number of crimes at the neighborhood level.²⁷ Consistent with multitasking crime response efforts reducing the marginal cost of ticketing, we find the strongest ticketing responses in areas with the greatest amount of criminal activity, which subsequently concentrates ticketing in Black neighborhoods. In stark contrast, the same correlation for non-CPD agents (Panel B) is flat, indicating no relationship between a neighborhood’s underlying crime level and the issuing of sticker tickets.

We further confirm that higher neighborhood crime levels are a key driver of this ticketing response in Table 3, where we formally test whether alternative neighborhood characteristics invoke similar differential responses. In Column 2, we define “high crime” neighborhoods as being in the top quartile of crime rates, finding strong evidence of differential ticketing in these neighborhoods. Alternatively, if officers are behaving in a purely revenue-maximizing way (without incorporating search costs), then sticker tickets should be written in the areas that have the highest repayment probabilities, such as high-income neighborhoods.²⁸ In fact, we find the opposite patterns in Column 3.

In Column 4, we show that CPD officers write more sticker tickets in areas that previously had high rates of sticker tickets (as a fraction of total tickets written in the neighborhood), suggesting that officers are aware of neighborhoods with low sticker compliance rates and alter their search effort accordingly. However, Column 5 shows that fewer sticker tickets are written in neighborhoods with high payment rates. When we test all interactions jointly, we still find that while the coefficient on neighborhood demographics remains large, disparities seem to also be significantly associated with area crime characteristics, further confirming that CPD officers increase ticketing behavior in areas where they are already located due to other responsibilities.²⁹ Alternative non-race and non-crime channels are generally less important.³⁰

²⁷Formally, we consider all one-way differences for $i \in \mathcal{I}$ and $R_i \in \{B, n\}$ of $E[Y_{it}|R_i = r, Post_t = 1] - E[Y_{it}|R_i = r, Post_t = 0]$, which represent each component piece of our difference-in-differences estimates. Averaging across all of these estimates by neighborhood type R_i and then taking the difference recovers our main DiD estimates.

²⁸It’s ambiguous whether officers would fully internalize potential ticket contesting or repayment probabilities when comparing the value of a ticket in high and low-income neighborhoods. Higher-income neighborhoods likely have more resources to fight parking tickets, but to the extent officers receive overtime pay for appearing in court (e.g., Chalfin and Goncalves, 2023), the marginal ticket in a high-income neighborhood becomes attractive both with respect to repayment probabilities as well as potential private value to the officer.

²⁹It is worth noting the tension of this finding with how CPD’s union describes the multi-tasking problem: “fewer tickets are written in higher-crime areas because officers are busy chasing more serious offenses” (<https://abc7chicago.com/archive/7614129/>).

³⁰We further test for differential officer behavior *within* Black neighborhoods by examining all alternative treatment margins above, but defined within the set of Black neighborhoods, as opposed to the overall sample in Appendix Table A4. Within Black neighborhoods, we find evidence that sticker tickets are weakly more likely to be written in higher-income and the highest crime areas. Consistent with the previous

Finally, we show that this passive multitasking response persists across the daily crime distribution. We correlate mean daily sticker tickets against daily crime incidents in a balanced zip code x date panel, separately for Black and non-Black neighborhoods, pre- and post-reform in Appendix Figure A8. Perhaps strikingly, average daily sticker tickets across the crime profile are similar between Black and non-Black neighborhoods prior to the reform. After the reform, however, while both neighborhood types experience an increase in their ticketing volume, Black neighborhoods experience a precipitous increase in sticker ticketing that persists across both overall daily crime distribution and when stratified by crime type, which we classify using the approach from Choi et al. (2023). Moreover, the rise of sticker ticketing in non-Black neighborhoods further highlights how increased neighborhood crime and police presence reduce the marginal search cost of ticketing.³¹

Alternatively, CPD officers may *dynamically multitask* and ticket in the immediate vicinity following a response to a criminal incident, rather than during slack time. We test to what extent CPD officers exhibit dynamic multitasking in Appendix Figure A10, where we plot coefficient estimates from event studies measuring the number of tickets written each hour within 300 meters of criminal incidents.³² While these estimates, especially when examining specific crime types are subject to noisier pre-trends, we find some evidence of differential dynamic multitasking in Black neighborhoods. Recovering the analogous difference-in-differences estimate by averaging over the post-incident indicators, our estimates suggest that the reform induced an additional 0.10 sticker tickets per hour in Black neighborhoods, during the four hours following a criminal incident. However, even after aggregating across mean number of criminal incidents, dynamic multitasking accounts for only around one-sixth of the overall ticketing gap (401/2,490), suggesting other external factors induced changes in multitasking priorities.³³

cross-neighborhood exercise, non-CPD agents exhibit little differential enforcement across subsets of Black neighborhoods.

³¹Further evidence of passive multitasking and incidental ticketing is in Appendix Figure A9, where we study how disparate enforcement varies by quartile of vehicle MSRP, using the average April 2025 MSRP by vehicle make, which requires an additional assumption of temporal cross-make ordinal price stability. We then plot the share of the overall DiD estimate by quartile, divided by the Black per-period share of vehicles in each quartile. Ratios above one indicate that there is greater enforcement in Black neighborhoods for those specific vehicle quartiles than would be expected if the enforcement was shared proportionally across vehicle price quartiles. Consistent with incidental ticketing and passive multitasking, we find that our estimates of disparate ticketing are concentrated among vehicles in the lowest price quartile, which may be more likely to be parked on the street or easily visible to ticketing officers.

³²A typical city block in Chicago is around 100 meters by 200 meters. Our estimated event studies include the full set of interactions between relative time indicators, neighborhood type, and reform period, while also controlling for zip code, crime type, and date-by-hour fixed effects. We then take linear combinations of the former set of interactions to form the event-study coefficients and use a 50 percent random sample of incidents to preserve computational tractability.

³³Specifically, we multiply the change in the ticketing gap (0.10) by the number of follow-up hours (4) and

Why would CPD’s multitasking behaviors change as a result of the reform? We provide two pieces of evidence that CPD officers are differentially responsive to revenue concerns, perhaps due to top-down revenue pressures and fiscal challenges (Lentino, 2017). First, in Figure 4, we partially test whether officers exert greater search effort into finding vehicles with expired stickers across all areas by plotting the number of sticker tickets issued by day for 2011 and 2012 in Figure 4. If officers exert greater search effort when the revenue benefit of doing so is greater, then we would expect to see increases in ticketing frequency immediately after the 15-day sticker renewal grace period ends and tickets can be legally issued. In Panel A, we see exactly this pattern for CPD, with a large spike in sticker tickets written on the day the grace period ends. Comparing the change between 2011 and 2012, sticker ticket volume increased by a dramatic 49.5% on the day immediately following the grace period.

In contrast, non-CPD agencies (Panel B) exhibit only a 17.0% increase in sticker ticket volume.³⁴ Interestingly, we also see small ticket volumes in the days preceding the end of the grace period, although an expired sticker should technically not be subject to enforcement in this window. The share of sticker tickets written in the expiration-grace period window is smaller for non-CPD agencies (<1%) than CPD agencies (between 4.9 - 5.6%), which we take as suggestive evidence of ex-ante ticket writing optimization by the former. The same histograms for non-sticker tickets have little evidence of similar discontinuous behavior.

Second, we provide additional evidence CPD officers are responsive to revenue collection objectives using the sale of Chicago parking meters to private investors in December of 2008. Under the privatization agreement, the city of Chicago would receive a one-time payment of \$1.15 billion from private investors in exchange for the next 75 years of parking meter revenue. Since the revenue from parking meter tickets no longer flows to the city budget after the sale, CPD officers should be less likely to write meter tickets, relative to non-CPD contractors, if CPD ticketing patterns are responsive to revenue incentives. We test this hypothesis in Appendix Figure A11 and find sharp relative decreases in CPD-written meter tickets after meter privatization, consistent with CPD officers altering their ticketing behavior in response to the destination of the revenue.³⁵ Taken together, we conclude that these revenue

the mean annual number of criminal incidents in Black neighborhoods in the post-period (around 10,000). As discussed above, differences in car accessibility could also be a driver of the differential dynamic sticker ticketing increase in Black neighborhoods.

³⁴The cyclical pattern in non-CPD ticket volume is a weekend/weekday effect. Therefore, we compare ticket volume on the first weekday after the grace period in 2011 (two days after it ends) against the initial expiration day in 2012 in the above calculation.

³⁵Consistent with general revenue concerns being a key motivator, we also find suggestive evidence of relative increases in non-sticker ticketing in Appendix Figure A12, however these estimates are partially based on unstable pre-trends.

considerations induced changes to the CPD’s ticketing behavior, which was differentially concentrated in Black neighborhoods due to lower marginal search costs, resulting in the disparate increase in sticker ticketing we document here.

Differential Compliance and Evasion: Concerns over tax evasion have a rich history in the public finance literature (Allingham and Sandmo, 1972; Slemrod and Yitzhaki, 2002), and one interpretation of our existing estimates is that they simply reflect differences in the willingness or ability of drivers to pay for the city sticker or evade enforcement efforts. As described above, differential ex-ante compliance with the policy may present itself as disparate ex-post enforcement of the sticker tax if the marginal benefit of writing such a ticket has increased, such that law enforcement agencies are now writing tickets they would not otherwise have in the absence of the budget reform. Alternatively, the 2012 budget reform may have changed compliance rates themselves since it also increased the price of the sticker for both small (\$75 to \$85) and large (\$120 to \$135) vehicles. If there is a sufficiently large subpopulation on the margin of sticker purchasing, then such an increase may lead to disparate enforcement as the marginal search cost of finding a delinquent motorist has relatively decreased in areas with larger decreases (or smaller increases) of sticker purchases. While we are unable to directly measure sticker purchases at the individual level due to data limitations, we conduct several tests to probe how much our estimates may reflect differential compliance versus differential enforcement of the policy.

First, we examine neighborhood-level sticker purchasing behavior using administrative data on sticker purchases, owner locations, and sticker types from 2008-2016. In Panel A of Appendix Figure A13, we plot event-study estimates of the interaction of year and neighborhood-type indicators, using sticker purchases as the outcome. If there were differential non-compliance with the sticker policy, such that the fine increase induced a large fraction of the non-complying population to suddenly purchase stickers, then we should see greater purchase rates in Black neighborhoods relative to non-Black neighborhoods. Alternatively, the sticker price increase may also lead to a differential reduction in purchasing as marginal individuals are priced out of compliance. If anything, Black neighborhoods have minuscule decreases (38 stickers) relative to non-Black neighborhoods, although the pre-trend estimates are somewhat noisy. However, in aggregate, both groups increased their purchases. When we disaggregate sticker purchases into types focusing on passenger vehicles, we find no statistically distinguishable differential response. This empirical finding is instead consistent with changes in officer behavior rather than substantial changes in civilian behavior due to price-out non-compliance or incentivized purchasing.³⁶

³⁶We confirm this finding through a decomposition exercise in Appendix Table A5 where we relate the magnitude of the enforcement and compliance responses across neighborhoods, adjusting for the presence of

Car owners may also alter their parking behavior in an attempt to evade enforcement. Enforcement may vary based on parking location.³⁷ We can (with some noise) measure whether a ticket was issued in a parking lot or structure. To do so, we measure whether a sticker ticket was issued at a location classified by Open Street Maps as a parking amenity.³⁸ 3.3% of CPD sticker tickets are issued in parking amenities (4.1% in non-Black zip codes, 2.2% in Black). 4.5% of non-CPD sticker tickets are issued in parking amenities (5.3% in non-Black zip codes, 1.6% in Black). We map this in Appendix Figure A2.

While not a perfect test for car owner behavior, Appendix Figure A14 shows that the share of tickets issued by CPD in parking areas does not substantially change in Black zip codes relative to non-Black zip codes after the policy change. This result is evidence that drivers in non-Black areas are not changing their parking behavior in order to avoid more costly enforcement and instead is consistent with increased search effort from CPD.³⁹

Is Differential Enforcement Maximizing Revenue?: Black neighborhoods receive substantially greater levels of sticker ticket enforcement, in magnitudes that cannot be rationalized by changes in sticker purchasing behavior. Moreover, this disparate effect is entirely driven by CPD rather than non-CPD ticketing agencies. One possible rationale for these empirical patterns is that CPD officers are differentially targeting areas with lower sticker purchase rates or higher repayment rates, while non-CPD officers have already equalized their marginal costs of enforcement across neighborhoods. To understand how neighborhood responses and baseline characteristics influence enforcement behavior, we construct joint distributions of estimated changes in neighborhood-level ticketing outcomes and baseline neighborhood characteristics.

In Panels A and B of Figure 5, we plot the joint distributions of neighborhood-level changes in sticker tickets and sticker purchases, separately by neighborhood type and ticket issuing agency. There is a weakly positive correlation between changes in sticker purchases and CPD-issued sticker tickets in non-Black neighborhoods, which is sharply contrasted with

unregistered vehicles. We continue to find strong differences in enforcement responses across neighborhoods alongside no detectable differences in compliance behavior or stark gaps in the stock of unregistered vehicles.

³⁷To the extent vehicles in Black neighborhoods are more likely to be parked on the street or visible to the average patrol (e.g., Sanchez and Ramos, 2018), then Black neighborhoods may see differential levels of sticker enforcement even in the pre-reform period since Black neighborhoods exhibit a lower marginal search cost for sticker-less vehicles.

³⁸See <https://wiki.openstreetmap.org/wiki/Tag:amenity=parking> for details on how Open Street Map classifies parking amenities. Mapping tickets into parking areas is done with some noise, as the ticket latitude and longitude are often for the street address rather than within the parking area. Thus, we impose a buffer, and a ticket is deemed “in a parking amenity” if it occurs within 15 meters from a parking amenity or within a parking amenity as defined by Open Street Mapping. Results are robust to alternative buffer regions; even at a buffer of 30m, fewer than 9% of sticker tickets are in parking amenities.

³⁹In fact, Appendix Figure A15 demonstrates that the share of sticker tickets issued near schools, grocers, parks, or residential roads is relatively stable.

a negative relationship in Black neighborhoods (Panel A). Strikingly, non-CPD-issued sticker tickets display parallel and virtually flat relationships across neighborhood types (Panel B). A possible interpretation of the negative slope is that CPD agents are more successfully capturing every marginal change in sticker purchases; perfect enforcement would suggest a slope of -1 (if each newly compliant vehicle would have received one ticket per period in the absence of compliance). We estimate a slope coefficient of -0.52. However, this slope difference does not fully explain the level difference between Black and non-Black neighborhoods with similar purchase responses, suggesting alternative mechanisms such as search costs.

We further explore whether differences in neighborhood characteristics can explain the CPD ticketing gap in Appendix Figure A16. In Panel A, we test whether differential pre-reform sticker purchase rates can rationalize such a gap, but we continue to find a persistent level difference, with virtually all neighborhood purchase rates clustered closely around one.⁴⁰ Alternatively, such a disparity may be justified by revenue collection motivations if the repayment probabilities are higher in Black versus non-Black neighborhoods. However we find that sticker ticket volumes are higher in Black neighborhoods, despite having lower pre-reform sticker ticket payment probabilities (Panel B). Finally, we show that changes in ticket volumes remain higher in Black neighborhoods across almost all values of sticker tickets and paid sticker tickets per sticker purchase (Panels C and D), indicating sharply distinct enforcement responses across areas with similar baseline enforcement propensities.

Taken together, we conclude that observable differences in neighborhood characteristics, compliance, and expected payment probabilities are insufficient to fully explain the CPD sticker ticketing gap across Black and non-Black neighborhoods. Instead, our results are consistent with a differential response to the fine increase by ticketing agency, combined with differential marginal search costs across neighborhoods. Therefore, the interaction of departmental incentives and search costs together plays a key role in determining disparate enforcement patterns across agencies.

Given the overlap in the support of non-race neighborhood characteristics, there are potentially significant revenue gains from equalizing enforcement across Black and non-Black neighborhoods. A back-of-the-envelope calculation leveraging the joint distributions of neighborhood characteristics and neighborhood-level ticketing responses suggests that the city could have raised an additional \$1.9 million in annual revenue (\$1.3 million after accounting for administrative costs of bankruptcy filings and dismissals) by applying average equal enforcement in low-compliance non-Black neighborhoods.⁴¹

⁴⁰Some of the x-axis dispersion is likely due to measurement error in the number of vehicles since we rely on aggregated survey-based measures when constructing this statistic.

⁴¹To perform this calculation, we first identify non-Black neighborhoods with lower purchase per vehicle

6 Estimating Officer-Specific Responses to the Reform

Our aggregate event study results reveal disproportionate issuance of sticker tickets across neighborhoods. An open question is whether this disparate behavior is department-wide or if it is concentrated among a handful of officers who are high-volume ticket-writers.⁴² We estimate a modified version of our difference-in-differences specification above to decompose the response to the policy reform across the officer distribution. Formally, we estimate:

$$Y_{ijlt} = \underbrace{\sum_{j \in J} \delta_j (Z_j \times Post_{lt})}_{\text{Officer-specific responses}} + \underbrace{Z_j}_{\text{Officer fixed effect}} + \mathbf{X}'_{it}\pi + \nu_{ijlt} \quad (3)$$

for neighborhood i , officer j , ticket l , and year t .

We control for year, unit, neighborhood, and officer fixed effects.⁴³ We study racial differences in officer ticketing by interacting Y_{ijlt} with indicators for neighborhood racial composition instead of placing the interaction term on the right-hand side.

The coefficients of interest are the interactions δ_j , which, conditional on the officer fixed effects, nonparametrically capture the within-officer response to the change in incentives induced by the fine increase. When estimated at the sticker ticket level, δ_j can be interpreted as the officer-specific outcome of the marginal ticket written in response to the policy.⁴⁴

6.1 Decomposing Outcomes of Marginal Sticker Tickets

Figure 6 reports the distributions of officer-specific policy responses. In Panel A, we plot bins of the officer-specific probability of writing a sticker ticket separately for Black and non-Black

rates than the minimum rate of Black neighborhoods. Intuitively, these are the neighborhoods where non-compliance is highest, and the marginal cost of enforcement is consequently lowest (abstracting from parking patterns). We then assign the mean Black neighborhood enforcement volume to these neighborhoods and assume the payment probability is equal to the pre-period payment rate. The mean payment rate for these focal non-Black neighborhoods is 59.5%, relative to 46.9% in Black neighborhoods.

⁴²Our ticketing data includes officer badge numbers. Because these identifiers can change, we build an officer badge crosswalk using data from OpenOversight. We combine this with data from the Invisible Institute to generate an officer-level data set with information on the number and type of tickets written, unit and employment history, complaints against the officer, and officer demographic information (race/ethnicity, sex, and age).

⁴³These fixed effects account for time-invariant differences in responsibilities and ticket writing potential across both unit assignments and geography. We depart slightly from our aggregate analysis and use tract (rather than zip) fixed effects and racial composition to more closely approximate officer beat assignments. These modifications allow us to identify racial differences in policy response within all officers rather than only among the subset of officers whose assignment happens to be near zip code boundaries.

⁴⁴We restrict the sample to officers who write at least 100 tickets in our sample period and write tickets both before and after 2012 to alleviate noise concerns. These restrictions drop only a handful of officers. Unfortunately, we cannot conduct a similar exercise for non-CPD units due to higher rates of turnover in those departments.

neighborhoods against the overall officer-specific sticker ticket response ($P(sticker|ticket, race)$ against $P(sticker | ticket)$). We report similar findings from Empirical Bayes-adjusted estimates in Appendix Figure A17. This exercise effectively decomposes the marginal sticker ticket response outcomes within each officer. Comparing the slopes of the race-specific against the overall response yields similar conclusions to our aggregate event studies above - officers responding to the policy reform are substantially more likely to write sticker tickets in Black neighborhoods than non-Black neighborhoods (0.641 vs 0.359). The magnitude of the overall sticker ticket response on the x-axis also provides evidence of a “first-stage” response to the policy, as greater than 65% of officers are more likely to write a sticker ticket post-reform, indicative of a coordinated response across the department.⁴⁵

Panel B examines the officer-specific revenue responses accruing from payment sticker tickets. The correlations between the race-specific and overall responses suggest that over 60% of the revenue originates from Black neighborhoods. Together with our decomposition results, these correlations suggest that the unequal enforcement of the sticker ticket also led to an unequal tax burden across neighborhoods.

In Panels C and D, we decompose the outcomes of the marginal sticker ticket separately for Black (Panel C) and non-Black (Panel D) neighborhoods. Specifically, we plot the correlations between $P(ticket\ outcome | sticker, race)$ against $P(sticker | race)$. There are striking differences in sticker ticket outcomes across race. Under 35% of sticker tickets written in Black neighborhoods end in payment, compared to non-Black sticker tickets, which are almost 15 p.p. more likely to end in payment. Sticker tickets in Black neighborhoods are significantly more likely to end in financial strain, consistent with our aggregate results above. Fully 45.3% of sticker tickets either receive a notice of non-payment (39.7%) or end in the driver filing for bankruptcy (5.6%). While sticker tickets in non-Black neighborhoods also only end in payment about half the time, there are fewer adverse financial outcomes (23.2% notice, 1.3% bankruptcy). Taken together, our results align with our aggregate analysis that the marginal sticker ticket is more likely to occur in Black neighborhoods, disproportionately generates revenue from this population, and leads to substantially worse financial outcomes compared to non-Black neighborhoods, and that these effects are widespread across the officer distribution.⁴⁶

⁴⁵A version of this exercise which captures both the extensive and intensive margin, reveals that over 85% of officers are responsive in a “first-stage” sense, providing additional support of behavioral changes to the policy reform across the officer distribution.

⁴⁶The disparate marginal revenue result in Panel B despite the lower payment rate in Panel C is likely driven by increases by racial differences in accrual and payment of late fees and extra penalties. Regressing the revenue received against the paid ticket probabilities reveals that the marginal paid Black sticker ticket generates \$278 of revenue compared to \$236 from non-Black sticker tickets. Tickets can generate revenue despite being unpaid due to quickly accruing late fees and interest. Moreover, tickets in Black neighborhoods are twice as likely to be both unpaid and have any positive payments, while also resulting in about \$26 in

6.2 Correlating Policy Responses with Officer-Level Observables

Finally, we explore whether the magnitude of the disparate sticker response is correlated with officer-level observables. While the majority of officer responses indicate that they are responsive to the sticker ticket fine increase in a first-stage sense, understanding whether officer characteristics are predictive of the magnitude of observed responses is important for characterizing how different types of officers react to incentive changes, as well as for designing department-level policies that may mitigate such adverse incentive responses.

A simple explanation for our officer-level decomposition is that the responses reflect the demographics of the unit to which they are assigned. That is, officers assigned to units with greater Black population shares should also exhibit larger Black ticketing responses. Conversely, officers assigned to units with smaller Black population shares should exhibit larger non-Black ticketing responses. We test whether this mechanism drives our results in Appendix Figure A18, regressing race-specific δ_j responses against their modal unit assignment’s Black share of the population.⁴⁷ Consistent with neighborhood demographics playing a key role in determining the responses in Black neighborhoods, we find a strong, positive correlation between estimates of $\delta_j(\textit{Sticker}, \textit{Black})$ and neighborhood demographics. In sharp contrast, however, we find a muted correlation between estimates of $\delta_j(\textit{Sticker}, \textit{Non} - \textit{Black})$ and neighborhood demographics, rather than a negative correlation, which a pure neighborhood characteristics story would predict.

Given this stark contrast, we next examine whether officer observables predict their policy responses in Table 4, controlling for modal unit fixed effects throughout.⁴⁸ In each successive column, we test a series of covariates before pooling them all together in Column 4. Black officers consistently have smaller $\delta_j(\textit{Sticker}, \textit{Black})$ responses, along with more experienced officers, though this latter correlation dissipates in Column 4. In contrast, officer characteristics are generally uncorrelated with $\delta_j(\textit{Sticker}, \textit{Non} - \textit{Black})$, with the exception of ticket volume. These estimates suggest some degree of differential leniency or search effort based on the interaction of officer race and neighborhood demographics. The magnitudes of the δ_j responses in Black neighborhoods are declining with experience, which is also indicative of early career officers perhaps being more responsive to revenue collection efforts in ways that disparately impact the population.

More generally, the combination of a strong policy response across the officer distribution and weak correlations with observable characteristics suggests that the empirical patterns we

additional revenue, on average.

⁴⁷We restrict the sample to police officers who are in patrol units so that we can correctly estimate the Black population share in the assignment, as well as examine responses of police officers who are most likely affected by the policy.

⁴⁸We show the correlations using the complete set of officers in Appendix Table A6.

find in this paper result from a broader goal of revenue generation and multitasking responsibilities on a department-wide level. Given the disparate impacts in the population that clearly hinge on ticketing agency, revenue collection as one responsibility of law enforcement agencies may benefit from specialization.

7 Conclusion

In this paper, we examine the role of policing on the distribution of tax burden on residents by exploiting the fine increase for vehicle registration non-compliance in Chicago. Using this sharp change in 2012, we showed that enforcement of this fine was indeed disproportionately distributed in the population, with Black neighborhoods experiencing far greater changes in their ticket volumes than non-Black neighborhoods.

Across enforcement agencies, we only find significant evidence of disparate enforcement when examining the ticketing behavior of police officers in the Chicago Police Department and not from tickets issued by parking enforcement agents. We show that CPD’s disparate response patterns cannot be fully explained by differences in baseline non-race neighborhood characteristics, nor justified on collection revenue generation grounds or differential non-compliance. Instead, we provide evidence that the combination of a multi-dimensional objective function (and higher crime rates in Black neighborhoods) with differential marginal enforcement costs by neighborhood drive these disparate responses. A top-down focus on revenue production only serves to exacerbate these enforcement behaviors.

Together, our results provide evidence that revenue generation in local governments may benefit from specialization across collection agencies as a mechanism to mitigate disparate impacts in the local population. While parking tickets, particularly sticker tickets, currently function as a form of regressive tax, cities can implement a more equitable and revenue-generating ticketing regime by altering the incentives of the ticketing agents or by shifting parking enforcement responsibility to only parking enforcement agents. Either of these could simultaneously achieve more equitable outcomes while also raising additional revenue.

Finally, our results document preliminary evidence of a direct relationship between disparate policing and downstream financial consequences. Notably, nearly 10% of the additional sticker tickets in our setting were unpaid and included as part of bankruptcy filings. Thus, policies addressing disparate policing behavior may also reduce racial disparities in socioeconomic outcomes.

References

- Aghion, Philippe and Jean Tirole.** 1997. “Formal and Real Authority in Organizations.” *Journal of Political Economy*, 105(1): 1–29.
- Allingham, Michael G. and Agnar Sandmo.** 1972. “Income tax evasion: A theoretical analysis.” *Journal of Public Economics*, 1(3-4): 323–338.
- Andriesen, Chris.** 2012. “Due process denied: Illinois’ new scheme to take your tax refund.” *Illinois Policy*.
- Avenancio-León, Carlos F. and Troup Howard.** 2022. “The assessment gap: Racial inequalities in property taxation.” *The Quarterly Journal of Economics*, 137(3): 1383–1434.
- Ba, Bocar, Patrick Bayer, Nayoung Rim, Roman Rivera, and Modibo Sidibé.** 2022. “Police Officer Assignment and Neighborhood Crime.” *NBER Working Paper No. 29243*.
- Brown, Dorothy A.** 2022. *The whiteness of wealth: How the tax system impoverishes Black Americans—and how we can fix it*. Crown.
- Callaway, Brantly and Pedro H.C. Sant’Anna.** 2021. “Difference-in-Differences with multiple time periods.” *Journal of Econometrics*, 225(2): 200–230.
- Callaway, Brantly, Andrew Goodman-Bacon, and Pedro H.C. Sant’Anna.** 2024. “Difference-in-differences with a Continuous Treatment.” *NBER Working Paper No. 32117*.
- Chalfin, Aaron and Felipe Goncalves.** 2023. “Professional Motivations in the Public Sector: Evidence from Police Officers.” *NBER Working Paper No. 31985*.
- Choi, Jay, David Kilmer, Michael Mueller-Smith, and Sema A. Taheri.** 2023. “Hierarchical approaches to Text-based Offense Classification.” *Science Advances*, 9(9): eabq8123.
- Civic Federation.** 2012. “2012 Consumer Tax Changes for Chicago Residents.”
- Davis, Jonathan, Samuel Norris, Jadon Schmitt, Yotam Shem-Tov, and Chelsea Strickland.** 2025. “Mobile Crisis Response Teams Support Better Policing: Evidence from CAHOOTS.” *NBER Working Paper No. 33761*.
- Dee, Thomas S. and James Pyne.** 2022. “A community response approach to mental health and substance sbuse crises reduced crime.” *Science Advances*, 8(23): eabm2106.
- Department of Human Resources.** 2023. “Code: 9161 Chicago Police Officer.” In *City of Chicago: Public Safety*.
- Digital Editors.** 2021. “New York City Made \$565 Million From Parking Violations in 1 Record-Breaking Year.” *MotorBiscuit*.

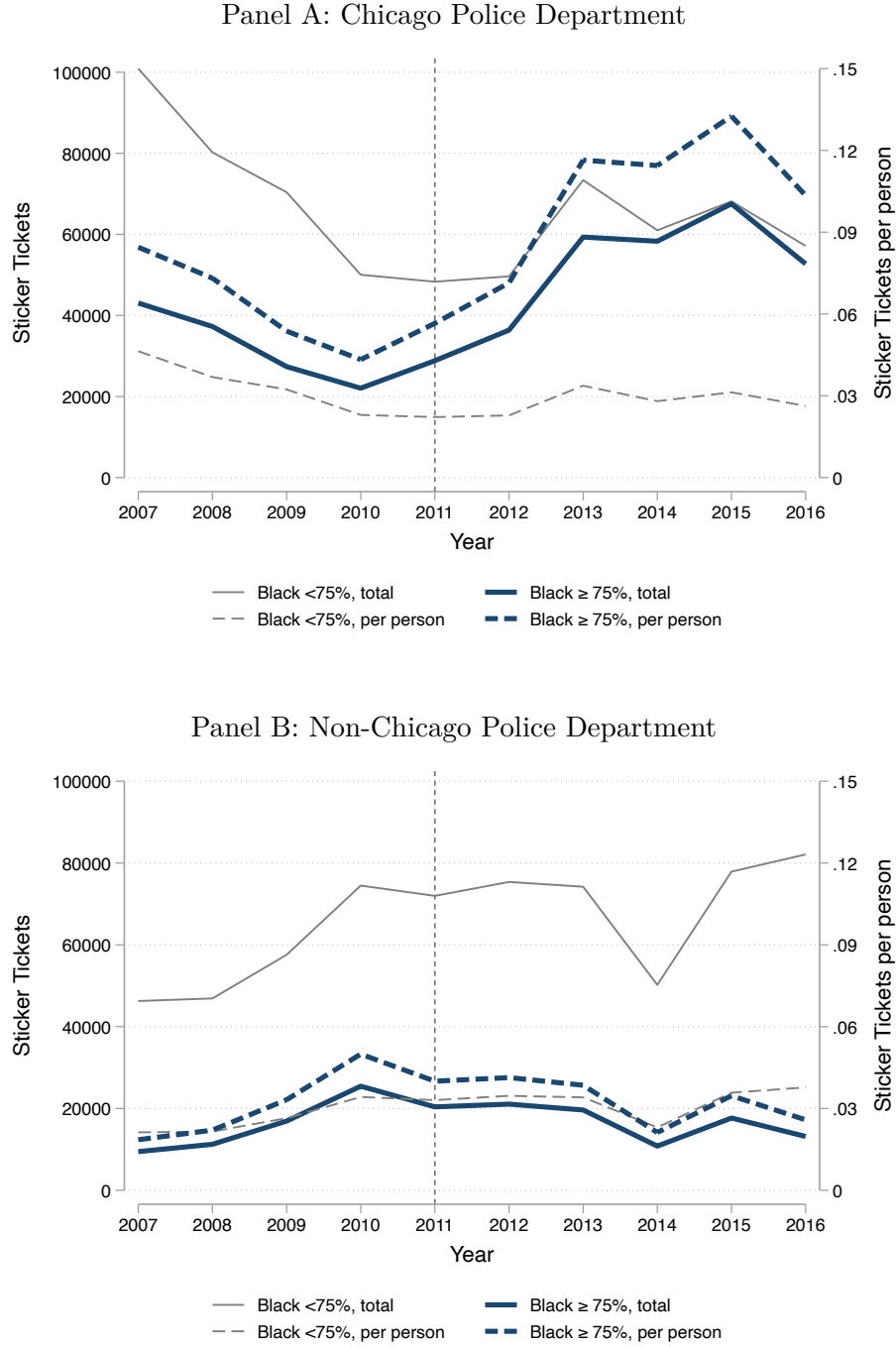
- Diskin, Eben.** 2019. “How much US cities make in parking ticket revenue will blow your mind.” *Matador Network*.
- Elias, Stephen.** 2007. *The New Bankruptcy: Will It Work For You?* NOLO.
- Elzayn, Hadi, Evelyn Smith, Thomas Hertz, Arun Ramesh, Jacob Goldin, Daniel E. Ho, and Robin Fisher.** 2025. “Measuring and mitigating racial disparities in tax audits.” *The Quarterly Journal of Economics*, 140(1): 113–163.
- Emanuel, Rahm.** 2011. “Mayor Rahm Emanuel outlines 2012 budget proposal to secure Chicago’s future.” *Mayor’s Press Office*.
- Finlay, Keith, Matthew Gross, Carl Lieberman, Elizabeth Luh, and Michael Mueller-Smith.** 2024. “The Impact of Criminal Financial Sanctions: A Multi-State Analysis of Survey and Administrative Data.” *American Economic Review: Insights*, 4: 490–508.
- Goldstein, Rebecca, Michael W. Sances, and Hye Young You.** 2020. “Exploitative Revenues, Law Enforcement, and the Quality of Government Service.” *Urban Affairs Review*, 56(1): 5–31.
- Goncalves, Felipe and Steven Mello.** 2021. “A Few Bad Apples? Racial Bias in Policing.” *American Economic Review*, 111(5): 1406–1441.
- Goncalves, Felipe and Steven Mello.** Forthcoming. “Police Discretion and Public Safety.” *Journal of Political Economy Microeconomics*.
- Goodman-Bacon, Andrew.** 2021. “Difference-in-differences with variation in treatment timing.” *Journal of Econometrics*, 225(2): 254–277.
- Goudie, Chuck, Barb Markoff, Christine Tressel, Tom Jones, and Maggie Green.** 2024. “Driven by Race: Chicago’s persistent problem of Black and white traffic stops.” *ABC7 Chicago*.
- Hansen, Benjamin.** 2015. “Punishment and Deterrence: Evidence from Drunk Driving.” *American Economic Review*, 105(4): 1581–1617.
- Harvey, Anna.** 2020. “Fiscal Incentives in Law Enforcement.” *American Law and Economics Review*, 22(1): 173–210.
- Holmstrom, Bengt and Paul Milgrom.** 1991. “Multitask Principal-Agent Analyses: Incentive Contracts, Asset Ownership, and Job Design.” *Journal of Law, Economics & Organization*, 7(Special Issue): 24–52.
- Holtzblatt, Janet, Swati Joshi, Nora R. Cahill, and William Gale.** 2023. “Racial Disparities in the Income Tax Treatment of Marriage.” *NBER Working Paper No. 31805*.
- Jacob, Brian A. and Steven D. Levitt.** 2003. “Rotten Apples: An Investigation of the Prevalence and Predictors of Teacher Cheating.” *The Quarterly Journal of Economics*, 118(3): 843–877.

- Kahn, Charles M., Emilson C. D. Silva, and James P. Ziliak.** 2001. “Performance-Based Wages in Tax Collection: The Brazilian Tax Collection Reform and Its Effects.” *The Economic Journal*, 111(468): 188–205.
- Kamenica, Emir.** 2012. “Behavioral Economics and Psychology of Incentives.” *Annual Review of Economics*, 4: 427–452.
- Kessler, Ryan E.** 2020. “Do Fines Cause Financial Distress? Evidence From Chicago.”
- Khan, Adnan Q., Asim I. Khwaja, and Benjamin A. Olken.** 2016. “Tax Farming Redux: Experimental Evidence on Performance Pay for Tax Collectors.” *The Quarterly Journal of Economics*, 131(1): 219–271.
- Knutsson, Daniel and Björn Tyrefors.** 2022. “The Quality and Efficiency of Public and Private Firms: Evidence from Ambulance Services.” *The Quarterly Journal of Economics*, 137(4): 2213–2262.
- Lazear, Edward P.** 2000. “Performance Pay and Productivity.” *American Economic Review*, 90(5): 1346–1361.
- Lentino, Chris.** 2017. “Chicago Looks to Add 50 ‘Super’ Ticket Writers to Increase Revenues.” *Illinois Policy*.
- Mahoney, Neale.** 2015. “Bankruptcy as Implicit Health Insurance.” *American Economic Review*, 105(2): 710–746.
- Main, Frank.** 2017. “ACLU blasts CPD commander for setting minimum number of stops.” *Chicago Sun Times*.
- Makowsky, Michael D. and Thomas Stratmann.** 2009. “Political Economy at Any Speed: What Determines Traffic Citations?” *American Economic Review*, 99(1): 509–527.
- Makowsky, Michael D. and Thomas Stratmann.** 2011. “More Tickets, Fewer Accidents: How Cash-Strapped Towns Make for Safer Roads.” *Journal of Law and Economics*, 54(4): 863–88.
- Makowsky, Michael D., Thomas Stratmann, and Alex Tabarrok.** 2019. “To Serve and Collect: The Fiscal and Racial Determinants of Law Enforcement.” *Journal of Legal Studies*, 48(1): 189–216.
- Manson, Steven, Jonathan Schroeder, David Van Riper, Katherine Knowle, Tracy Kugler, Finn Roberts, and Steven Ruggles.** 2024. “IPUMS National Historical Geographic Information System: Version 19.0 [dataset].”
- Mas, Alexandre.** 2006. “Pay, Reference Points, and Police Performance.” *The Quarterly Journal of Economics*, 121(3): 783–821.
- Mello, Steven.** 2025. “Fines and Financial Wellbeing.” *The Review of Economic Studies*, 92(5): 3340–3374.

- Morrison, Edward R., Belisa Pang, and Antoine Uettwiller.** 2020. "Race and bankruptcy: Explaining racial disparities in consumer bankruptcy." *The Journal of Law and Economics*, 63(2): 269–295.
- National Center for State Courts.** 2005. "Trial Court Performance Measures: Cost per Case."
- Owens, Emily and Bocar Ba.** 2021. "The Economics of Policing and Public Safety." *Journal of Economic Perspectives*, 35(4): 3–28.
- Pager, Devah, Rebecca Goldstein, Helen Ho, and Bruce Western.** 2022. "Criminalizing Poverty: The Consequences of Court Fees in a Randomized Experiment." *American Sociological Review*.
- Prendergast, Canice.** 2007. "The Motivation and Bias of Bureaucrats." *American Economic Review*, 97(1): 180–196.
- Prendergast, Canice.** 2008. "Intrinsic Motivation and Incentives." *American Economic Review: Papers & Proceedings*, 98(2): 201–205.
- ProsperityNow.** 2017. "The Racial Wealth Divide in Chicago."
- Ramos, Elliott and Melissa Sanchez.** 2018. "Chicago Hoped to Generate Millions with Expensive City Sticker Tickets. It Didn't Work." *WBEZ Chicago*.
- Reeves, James.** 2024. "Multitasking, Incentives, and Police Officer Behavior." *Working Paper*.
- Rivera, Susan.** 2010. "Are Chicago police being asked to make ticket quotas?" *NBC Chicago*.
- Ruthhart, Bill and Tribune Reporter.** 2014. "Debts to Chicago by its workers balloon." *Chicago Tribune*.
- Samuelson, Tracey.** 2018. "Parking tickets are bankrupting Chicago residents." *Marketplace*.
- Sanchez, Melissa and Elliot Ramos.** 2015. "Chicago Throws Out 23,000 Duplicate Tickets Issued Since 1992 to Motorists Who Didn't Have Vehicle Stickers." *ProPublica*.
- Sanchez, Melissa and Elliot Ramos.** 2018. "Chicago Hiked the Cost of Vehicle City Sticker Violations to Boost Revenue. But It's Driven More Low-Income, Black Motorists Into Debt." *ProPublica*.
- Sanchez, Melissa and Sandhya Kambhampati.** 2018. "How Chicago Ticket Debt Sends Black Motorists Into Bankruptcy." *ProPublica*.
- Sharkey, Patrick and Alisabeth Marsteller.** 2022. "Neighborhood Inequality and Violence in Chicago, 1965-2020." *The University of Chicago Law Review*, 89(2): 349–381.

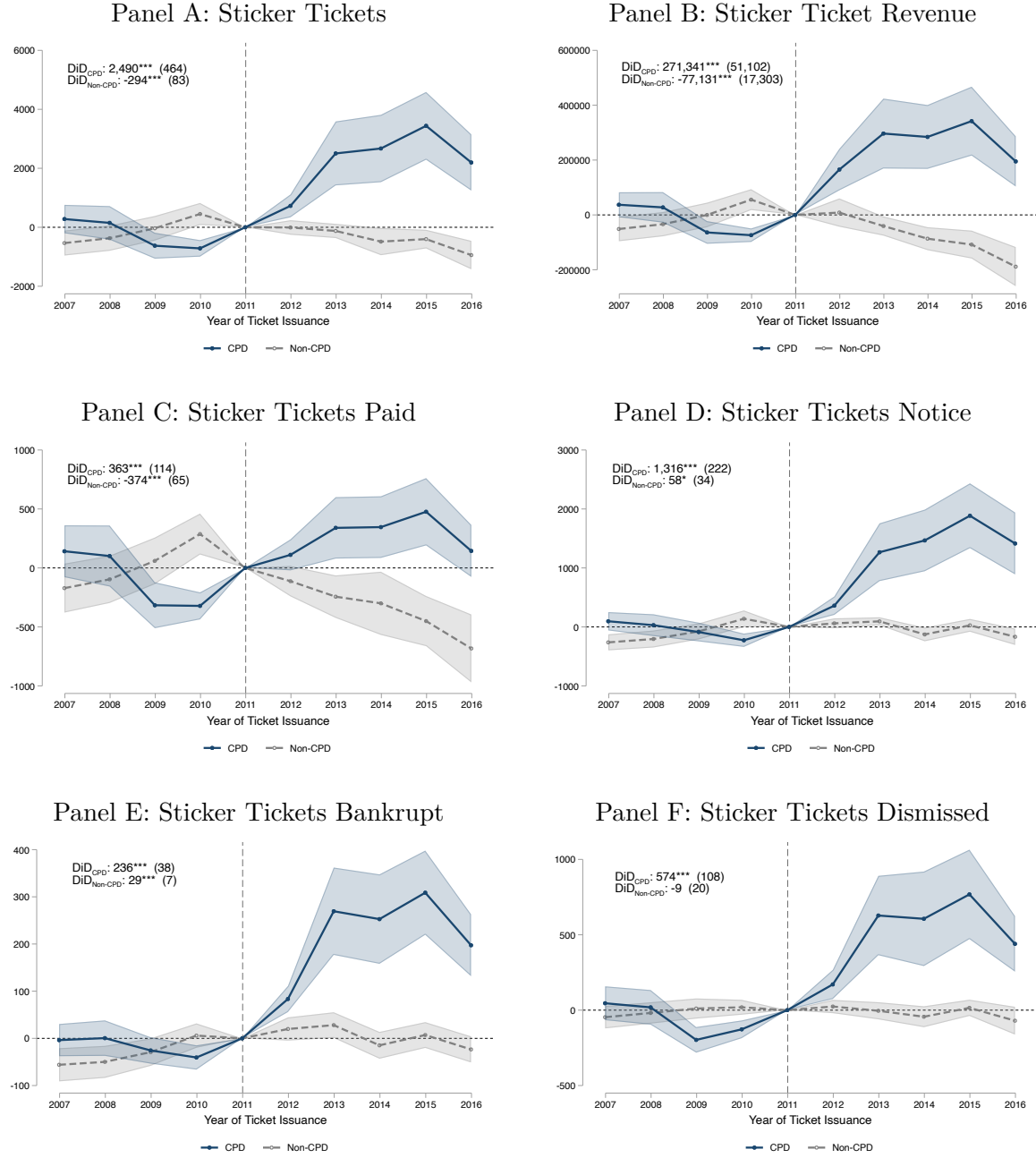
- Slemrod, Joel and Shlomo Yitzhaki.** 2002. “Chapter 22 - Tax Avoidance, Evasion, and Administration.” *Handbook of Public Economics*, 3: 1423–1470.
- Spielman, Fran and Tom Schuba.** 2024. “Johnson cancels 2 months of police academy classes, orders layoff lists to cut \$75M.” *The Chicago-Sun Times*.
- U.S. Department of Justice Civil Rights Division.** 2015. “Investigation of the Ferguson Police Department.” U.S. Department of Justice.

Figure 1: Time Series of Sticker Ticket Volume by Issuing Agency



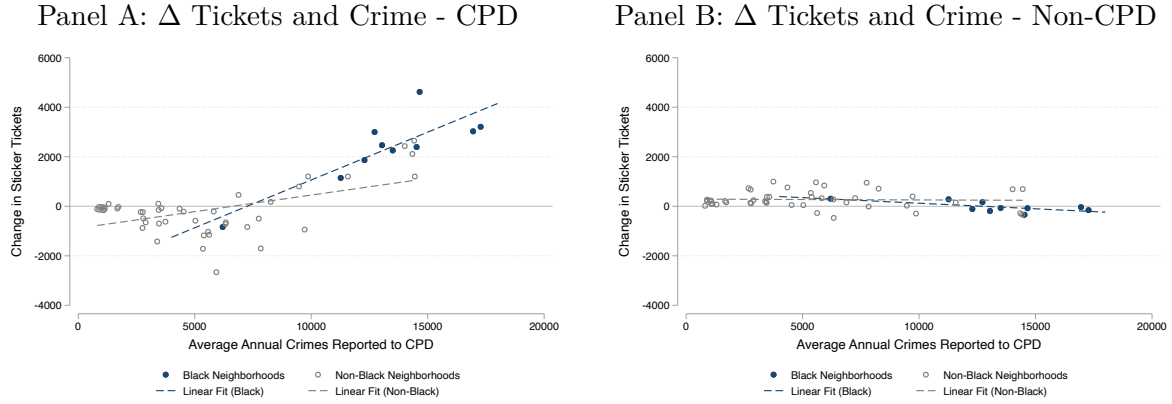
Notes: This figure reports time series of sticker tickets issued and sticker tickets issued per capita by neighborhood and ticketing agency. Black neighborhoods are defined as zip codes with greater than seventy-five percent Black population share. Panel A reports results for CPD and Panel B reports results for non-CPD agencies. Solid lines report levels (left axis), dashed lines report per-capita population rates (right axis). Blue lines represent Black neighborhoods and gray lines represent non-Black neighborhoods. The vertical line in 2011 denotes the last year prior to the reform. Population is measured using the 2007-2011 American Community Survey.

Figure 2: Event Study Estimates of Sticker Tickets and Outcomes at the Neighborhood-Level by Issuing Agency



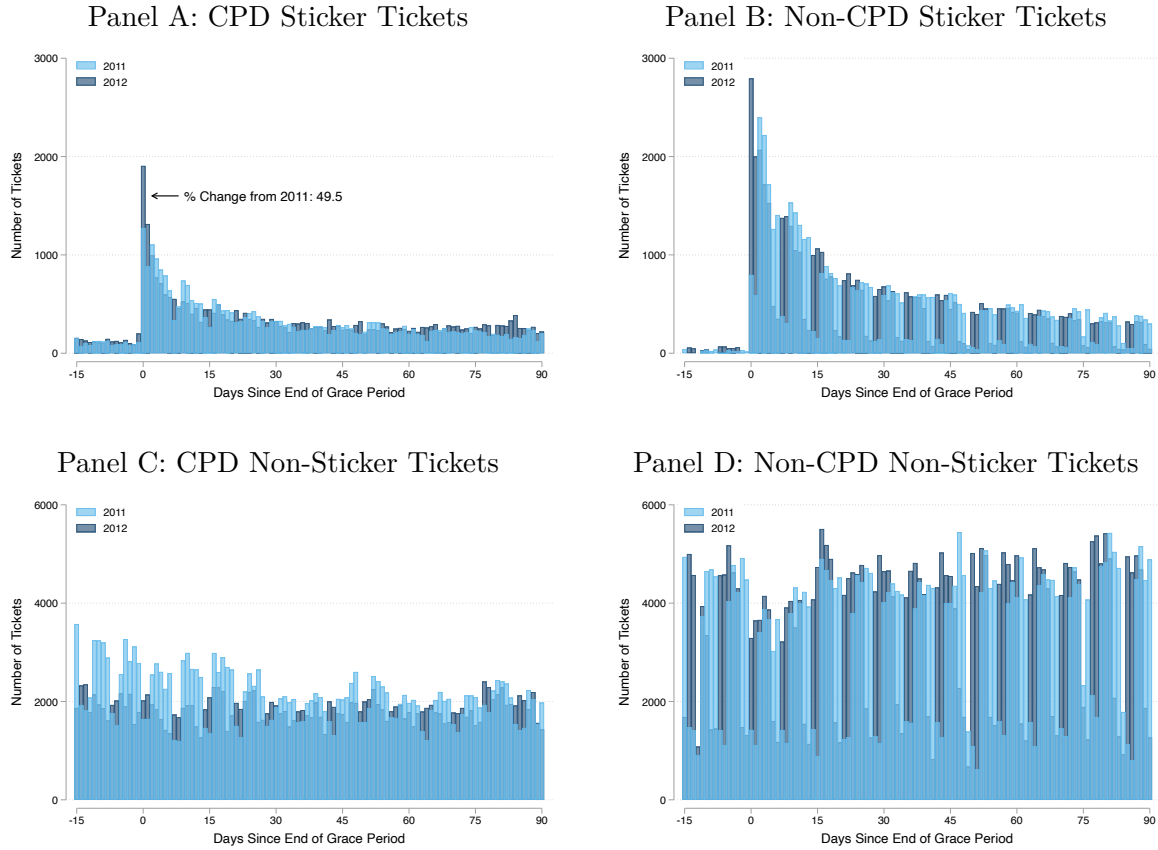
Notes: This figure reports event study estimates of sticker ticketing behavior and sticker ticket outcomes at the neighborhood level, estimated separately for the Chicago Police Department (CPD) and non-CPD agencies. Each point represents the interaction of $Black_i$ and the corresponding year fixed effect, relative to the level in 2011. The blue points report estimates for CPD and the gray points represent estimates for non-CPD. Shaded regions represent 95 percent confidence intervals with standard errors clustered at the neighborhood level.

Figure 3: Neighborhoods with Higher Crime Experience More CPD Ticketing



Notes: Notes: This figure reports joint distributions of neighborhood-level one-way difference estimates of the change in sticker ticketing, along with neighborhood-level estimates with pre-reform crime levels. Each panel plots the change in sticker tickets and neighborhood crime in levels for Black and non-Black neighborhoods, by CPD (Panel A) and non-CPD (Panel B) ticketing agency, respectively. Each point represents a neighborhood, defined at the zip code level. Navy dots represent Black neighborhoods and gray circles represent non-Black neighborhoods. Dashed lines represent linear lines of best fit, estimated separately by neighborhood type.

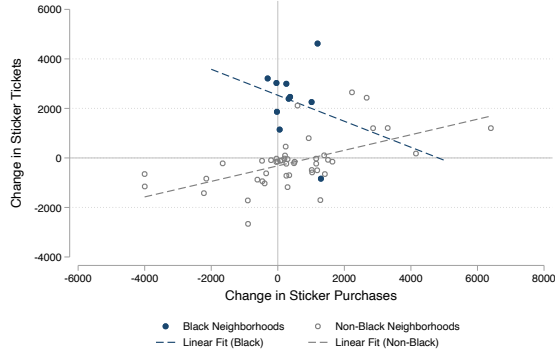
Figure 4: Distribution of Tickets Around the End of Sticker Renewal Grace Period:
2011-2012



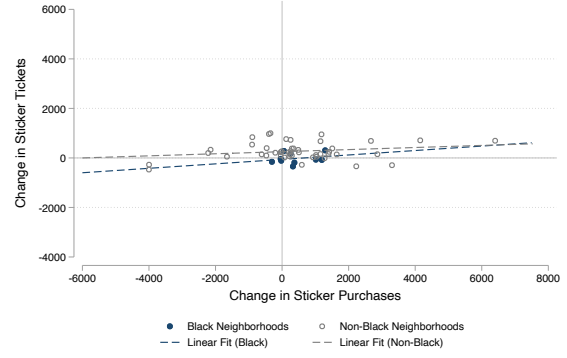
Notes: This figure reports the number of tickets issued per day in 2011 and 2012 by CPD and non-CPD ticketing agencies. Light blue histograms represent 2011 and dark blue histograms represent 2012. The x-axis is normalized to the end of the year-specific sticker renewal grace period and includes 15 days before and 90 days after the end of the grace period. Panels A and C ticketing distributions for CPD and Panels B and D report ticketing distributions for non-CPD agencies. The upper panels report the distribution of sticker tickets and the lower panels report the distribution of non-sticker tickets. The grace period is a window after the expiration date where an individual may purchase a city sticker without paying late fees and is not supposed to be subject to sticker ticket enforcement. We use 2011-2012 as the focal years in this exercise since, prior to 2014, all city stickers expired at the end of June in any given calendar year. The city shifted to time-varying city sticker expiration dates in 2014 when city stickers then expired 6 months after state-level car registration. Moreover, both 2011 and 2012 have the same fifteen-day grace period, which enables us to more cleanly harmonize and compare the data across years.

Figure 5: Relationship Between Change in Sticker Ticketing and Sticker Purchasing

Panel A: Δ Tickets and Δ Purchases - CPD

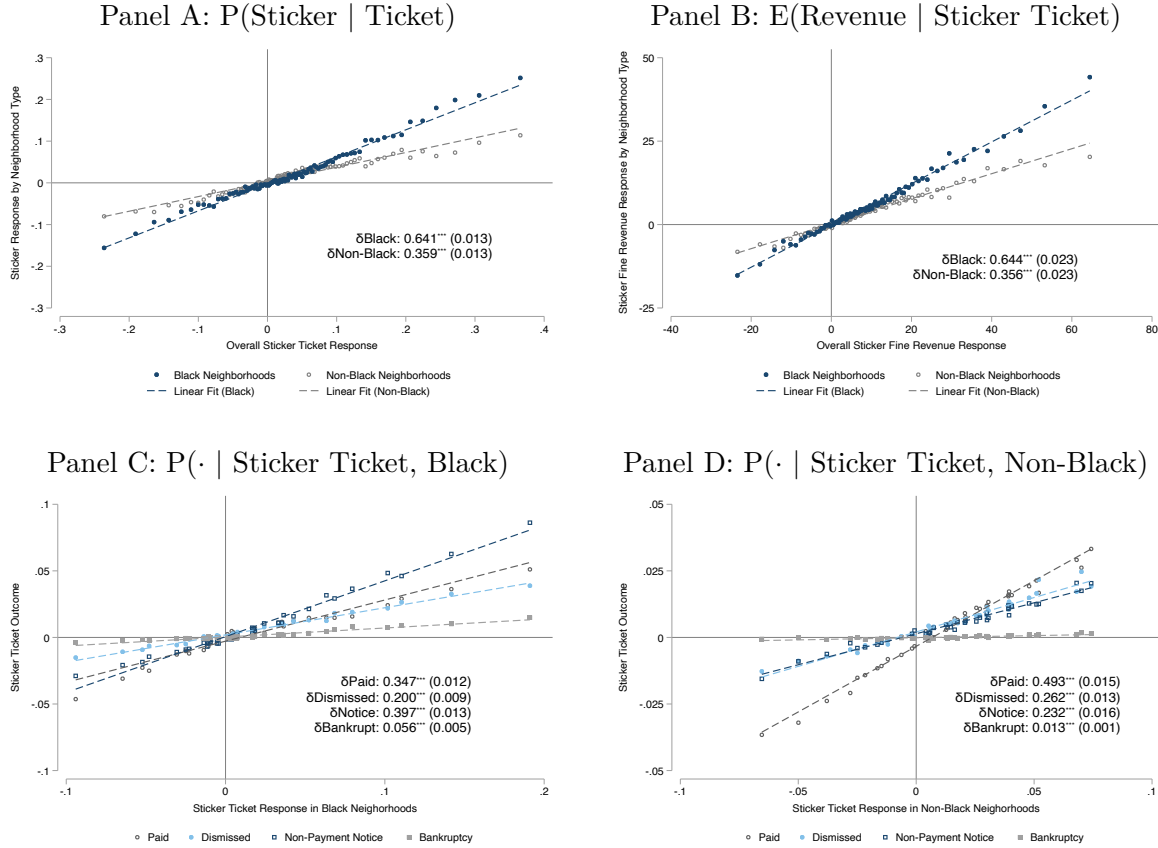


Panel B: Δ Tickets and Δ Purchases - Non-CPD



Notes: This figure reports joint distributions of neighborhood-level one-way difference estimates across different outcomes. Each panel plots the change in sticker tickets and sticker purchases for Black and non-Black neighborhoods, by CPD (Panel A) and non-CPD (Panel B) ticketing agency, respectively. Each point represents a neighborhood, defined at the zip code level. Navy dots represent Black neighborhoods and gray circles represent non-Black neighborhoods. Dashed lines represent linear lines of best fit, estimated separately by neighborhood type.

Figure 6: Estimating and Decomposing Officer-Specific Responses to Sticker Fine Increase



Notes: This figure plots estimates of δ_j for different outcomes against each other. In Panels A and B we plot estimates of neighborhood race-specific δ_j responses against the overall race-agnostic δ_j response on the x-axis. In Panels C and D, we plot the race-specific sticker ticket outcomes against the race-specific sticker ticket responses, separately by race. There are 100 bins per outcome in the top two panels and 40 bins per outcome in the lower two panels. For exposition, we drop the first and last bin for each outcome. Reported coefficients estimated on the underlying officer-level estimates. Robust standard errors are reported in parentheses. *** = significant at 1 percent level, ** = significant at 5 percent level, * = significant at 10 percent level.

Table 1: Annual Volume and Revenue from Top Ten Revenue Generating Tickets in Chicago

	2007-2011						2012-2016					
	Tickets	Revenue	Avg.	Rev.	Rev.	Rev./	Tickets	Revenue	Avg.	Rev.	Rev.	Rev./
	(000s)	(\$000s)	Share	Rank	Rank	E[Rev.]	(000s)	(\$000s)	Fine	Share	Rank	E[Rev.]
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Expired Meter	404	19,533	50	0.14	1	0.97	283	15,265	50	0.10	4	1.08
No City Sticker	178	18,614	120	0.14	2	0.87	205	29,575	200	0.20	1	0.72
Expired Plates	368	17,151	50	0.13	3	0.93	361	19,562	60	0.13	2	0.90
Street Cleaning	282	14,820	50	0.11	4	1.05	271	15,766	60	0.11	3	0.97
Residential Permit Parking	222	12,858	60	0.09	5	0.96	179	13,033	75	0.09	5	0.97
Parking Prohibited Anytime	145	8,046	60	0.06	6	0.93	129	9,270	75	0.06	6	0.96
Rush Hour Parking	122	7,123	60	0.05	7	0.97	67	5,533	100	0.04	8	0.82
Rear And Front Plate Required	154	6,782	50	0.05	8	0.88	117	4,622	60	0.03	10	0.66
Park In Transit Stand	51	4,906	100	0.04	9	0.95	38	3,589	100	0.02	11	0.95
Within 15' Of Fire Hydrant	46	4,549	100	0.03	10	0.99	43	5,119	150	0.03	9	0.80

Notes: This table reports average annual characteristics of the top ten revenue generating tickets from 2007-2011 and characteristics for the same set of tickets from 2012-2016. The type of ticket is listed in each row. Columns 1 and 7 report annual ticket volume, Columns 2 and 8 report annual ticket revenue, Columns 3 and 9 report the modal fine amount, Columns 4 and 10 report the revenue share of the listed ticket, along with the revenue share rank in Columns 5 and 11. Columns 6 and 12 report the ratio of revenue received and expected revenue, calculated as the base fine amount times ticket volume. The revenue share is less than one when collected revenue plus applicable late fees is less than the expected collected amount if all written tickets were paid on time, and is greater than one if collected revenue plus applicable late fees exceeds this expectation.

Table 2: Descriptive Statistics by Neighborhood: 2007-2011

	All Neighborhoods	Black Neighborhoods	Non-Black Neighborhoods
	(1)	(2)	(3)
<i>Panel A: Ticket-Level Outcomes</i>			
CPD Written	0.503	0.602	0.487
Sticker Ticket	0.076	0.149	0.064
Paid	0.527	0.471	0.548
Non-Payment Notice	0.228	0.308	0.197
Bankruptcy	0.021	0.039	0.014
Dismissed	0.224	0.182	0.240
CPD Written	0.572	0.655	0.541
<i>Panel B: Neighborhood-Level Outcomes</i>			
CPD Written	21,020	19,519	21,346
Sticker Ticket	3,176	4,842	2,814
Paid	1,674	2,280	1,542
Non-Payment Notice	723	1,492	556
Bankruptcy	66	188	40
Dismissed	713	882	676
CPD Written	1,816	3,173	1,521
Stickers Purchased	20,301	16,409	21,147
<i>Panel C: Neighborhood Characteristics</i>			
Share Black	0.305	0.934	0.168
Total Population	47,966	50,965	47,314
Total Vehicles	20,425	17,320	21,100

Notes: This table reports descriptive statistics by neighborhood type over the period 2007-2011. Panel A reports mean outcomes at the ticket-level and Panels B and C report mean annual outcomes at the neighborhood level. Column 1 reports overall means, Column 2 reports means in Black neighborhoods, defined as zip codes with a greater than seventy-five percent Black population share, and Column 3 reports means in non-Black neighborhoods. Sticker purchase data covers the period 2008-2011. Outcomes in Panel C are calculated using the 2007-2011 American Community Survey. We approximate total vehicles by aggregating bins based on survey responses and top code the highest bin as representing four vehicles.

Table 3: Testing Departmental Responses to Alternative Treatment Margins

	Sticker Tickets					
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: CPD</i>						
Black $\times$ Post	2,490*** (464)					1,129 (720)
High Crime $\times$ Post		2,005*** (426)				1,206** (523)
High Income $\times$ Post			-689** (268)			-328 (281)
High Sticker Ticket Rate $\times$ Post				1,755*** (460)		691* (369)
High Sticker Ticket Payment Rate $\times$ Post					-863*** (275)	30 (268)
<i>Panel B: Non-CPD</i>						
Black $\times$ Post	-294*** (83)					-40 (211)
High Crime $\times$ Post		-256*** (76)				-229 (141)
High Income $\times$ Post			80 (117)			139 (125)
High Sticker Ticket Rate $\times$ Post				-260*** (75)		-175 (115)
High Sticker Ticket Payment Rate $\times$ Post					-11 (106)	-189* (101)

Notes: This table presents difference-in-differences estimates using alternative treatment definitions. Panel A reports results for CPD-written tickets, and Panel B reports results for non-CPD-written tickets. The corresponding interaction is listed in each row. The outcome in all columns is the number of sticker tickets. Non-Black alternative treatment definitions are defined as being in the upper quartile or not. All regressions include zip code and year fixed effects. Standard errors clustered at the zip code level are reported in parentheses. *** = significant at 1 percent level, ** = significant at 5 percent level, * = significant at 10 percent level.

Table 4: Correlating Officer Policy Responses with Observable Characteristics

	δ_j Response			
	(1)	(2)	(3)	(4)
<i>Panel A: δ_j (Sticker, Black)</i>				
Male	-0.000 (0.003)			-0.000 (0.003)
Age	-0.000** (0.000)			-0.000 (0.000)
Hispanic		-0.002 (0.003)		-0.002 (0.003)
Asian or Native American		-0.001 (0.006)		-0.002 (0.006)
Black		-0.009** (0.004)		-0.008* (0.004)
Years Experience			-0.001*** (0.000)	-0.001 (0.000)
Complaints per Year			-0.000 (0.002)	-0.001 (0.002)
Tickets Issued per Year (00s)			0.002** (0.001)	0.001* (0.001)
<i>Panel B: δ_j (Sticker, Non-Black)</i>				
Male	-0.000 (0.002)			-0.001 (0.002)
Age	-0.000 (0.000)			-0.000 (0.000)
Hispanic		0.002 (0.002)		0.002 (0.002)
Asian or Native American		0.010 (0.006)		0.010* (0.006)
Black		0.001 (0.002)		0.002 (0.002)
Years Experience			0.000 (0.000)	0.000 (0.000)
Complaints per Year			0.001 (0.002)	0.001 (0.002)
Tickets Issued per Year (00s)			0.001** (0.001)	0.001** (0.001)
Observations	4,992	4,992	4,992	4,992
Unit Fixed Effects	Yes	Yes	Yes	Yes

Notes: This table reports regressions of officer-specific δ_j responses against officer-level observables. The sample includes only police officers in patrol units. The dependent variable in Panel A is the officer-specific δ_j for sticker tickets in Black neighborhoods and the dependent variable in Panel B is the corresponding δ_j for sticker tickets in Non-Black neighborhoods. Experience, complaints and tickets issued per year are all measured prior to the policy change. Robust standard errors are reported in parentheses. *** = significant at 1 percent level, ** = significant at 5 percent level, * = significant at 10 percent level.